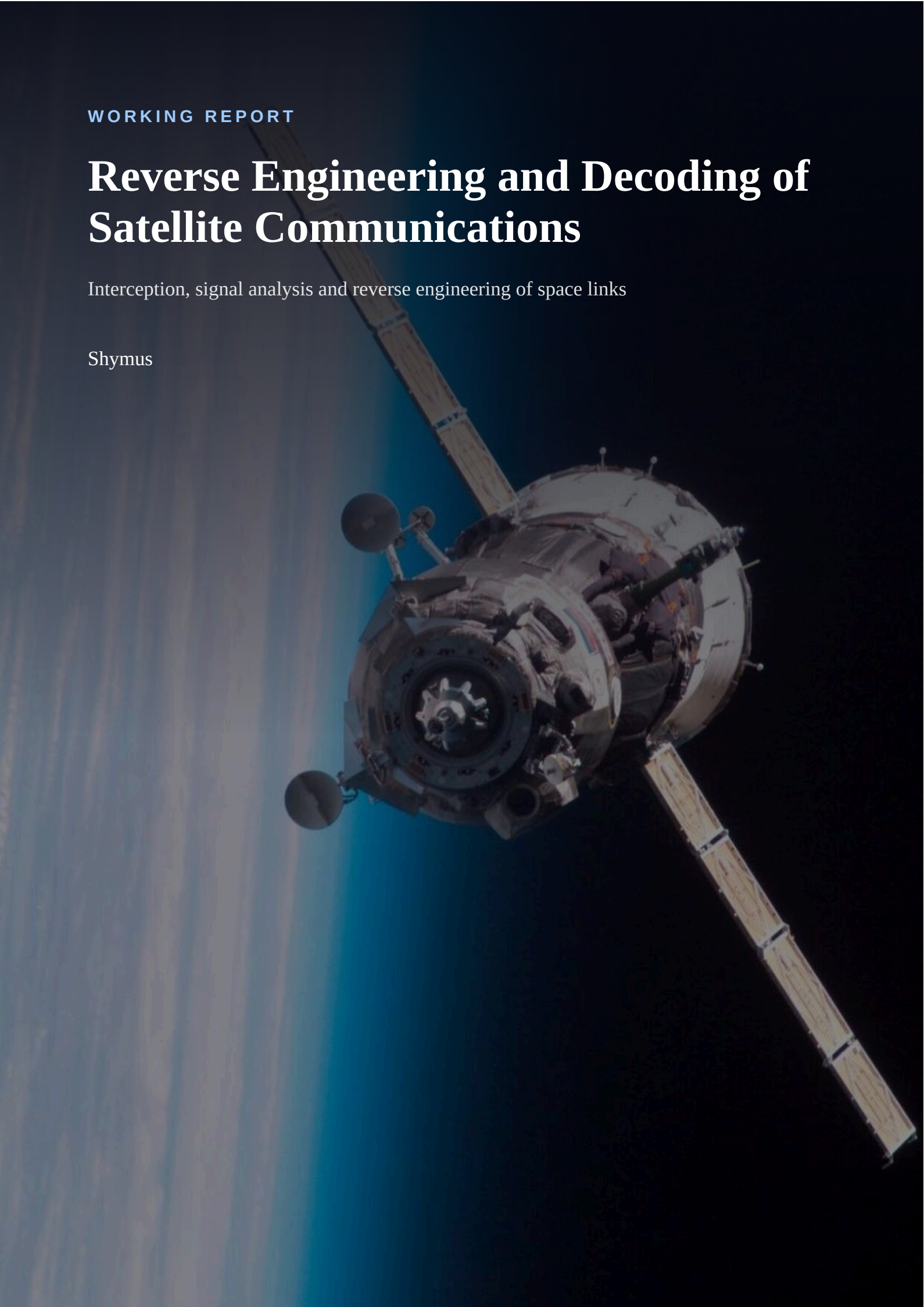
WORKING REPORT

Reverse Engineering and Decoding of Satellite Communications

Interception, signal analysis and reverse engineering of space links

Shymus



Foreword

This document is a working report devoted to the reverse engineering and decoding of satellite communications, from an aerospace-cybersecurity perspective. It documents the full set of hands-on work I carried out around satellite reverse engineering and decoding, across four parts: (1) an overview of satellite communications; (2) analysis with gpredict, SatDump and WXtoIMG; (3) decoding with Wireshark and GNU Radio; (4) SatNOGS and reverse engineering. For each part, this report covers the concepts, the tools installed and the labs performed, illustrated by the corresponding screenshots, then puts them back into context using authoritative external references (CCSDS, AMSAT, SatNOGS, SatDump, GNU Radio, Wireshark, and others).

Writing conventions. Established technical terms are kept in their original form (downlink, transponder, fuzzing, baseband frame, and so on).

Ethical and legal notice. Every technique described here concerns the passive reception and analysis of freely accessible signals (weather satellites, amateur-radio satellites, unencrypted DVB broadcasts), as well as teaching exercises (CTF challenges, test firmware). Rules on listening, radio transmission and the interception of communications vary from country to country; this document is strictly educational and defensive in purpose.

CHAPTER 1

Introduction and methodology

1.1 Context and motivation

Satellite communications are today one of the most accessible playing fields in cybersecurity, because it is possible to study them ethically with cheap hardware and free software. The radio link between a satellite and the ground is, by nature, a broadcast channel: any station equipped with a suitable antenna and a receiver can passively listen to the signals falling back from space. That property makes the communication link the prime entry point for understanding the security of space systems.

The tooling ecosystem is almost entirely open source and free: software-defined radios (SDR) for a few tens of euros, orbit-prediction software, image decoders, packet analysers and reverse-engineering tools. This report walks through the whole chain, from gathering orbital telemetry to reverse-engineering frames and firmware.

1.2 Scope: listening, interception and reconnaissance

The scope covers three closely linked families of scenarios: the eavesdropping of signals, the interception of communications, and the reconnaissance that precedes a contact. These activities rely on public data sources (orbital parameters, transponder datasheets) and on consumer-grade receiving hardware.

1.3 The four layers of analysis

Analysing a satellite communication can be broken down into four successive layers, which structure this whole document:

1. **Orbital telemetry gathering:** actively determine the satellite's position and orbital parameters, its band and its frequencies, using online data sources and pass-prediction software.
2. **Listening to the satellite:** choose and aim the right antenna, configure a receiver (handheld, SDR) and record the signal.
3. **Packet analysis and dissection:** extract realistic frames, identify the protocols used (CCSDS, DVB, AX.25, CSP…) and dissect their content.
4. **Reverse engineering and firmware auditing:** turn waveforms into digital data, reconstruct the protocols, then audit embedded firmware by fuzzing.

These four layers form a coherent pipeline: each step produces the information the next one needs.

CHAPTER 2

Satellite communication fundamentals

Before any interception, you have to understand the receiving equipment and the physics at play. This chapter lays the groundwork: antennas, polarisation, transponders, the Doppler effect, operating modes and software radios.

2.1 Gearing up to listen: handheld radio and Yagi antenna

Anyone can listen to a satellite with very little equipment. To get started, two items are enough: a dual-band handheld transceiver (HT), and a portable Yagi antenna (directional). For FM satellites, the radio should ideally support full duplex (transmitting and receiving at the same time), as opposed to half duplex where only one party talks at a time.

Classic entry-level models include the Kenwood TH-D7A or, more affordably (around $30–40), the dual-band Baofeng UV-5R. A manual Yagi antenna (~$20) such as the HYS Yagi (UHF 400–470 MHz) is enough to pick up the International Space Station (ISS). The total cost stays under a hundred dollars. AMSAT's “For Beginners” resource [23] is an excellent gateway to amateur-radio hardware.

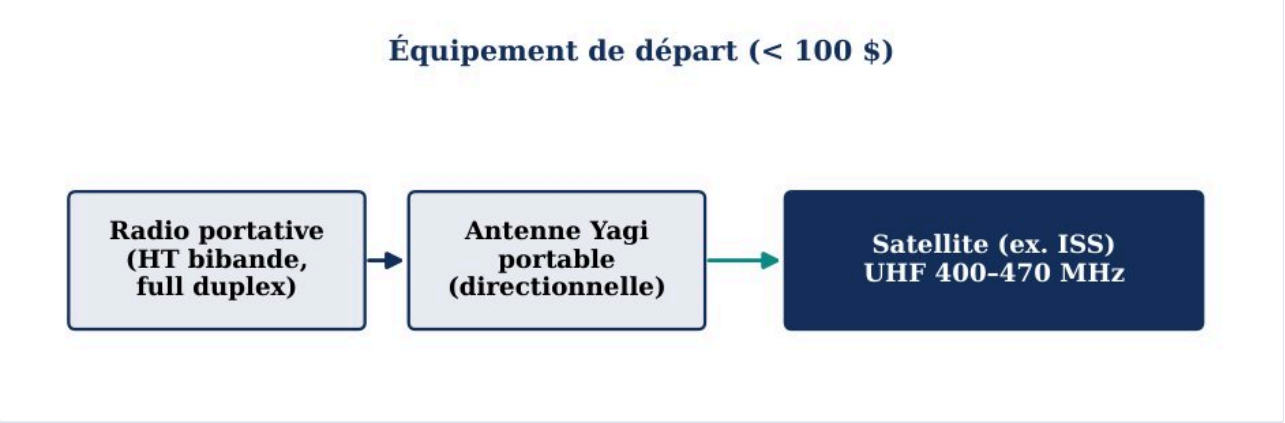


Figure 2.1 — Minimal gear for starting to listen to satellites (a dual-band handheld and a Yagi antenna).

2.2 Antenna types: directional vs omnidirectional

Directional antennas concentrate RF energy in a precise direction. Advantages: greater range and the ability to catch weak signals. Drawback: they must be aimed correctly, or you receive nothing. The loss between a poorly set vertical and horizontal polarisation can reach 30 dB.



Figure 2.2 — Some illustrations come from public videos; here, the timestamped source frame of a listening demonstration using a Yagi antenna.

Omnidirectional antennas are more forgiving and better suited to beginners. Circularly polarised, they allow only two configurations (RHCP, right-hand, and LHCP, left-hand), with a loss of only 3 dB in case of error. Examples: Lindenblad, turnstile, quadrifilar helix (QFH). The choice is a trade-off: the omnidirectional avoids fine tuning but sacrifices signal strength; the directional offers better quality at the price of rigorous pointing.

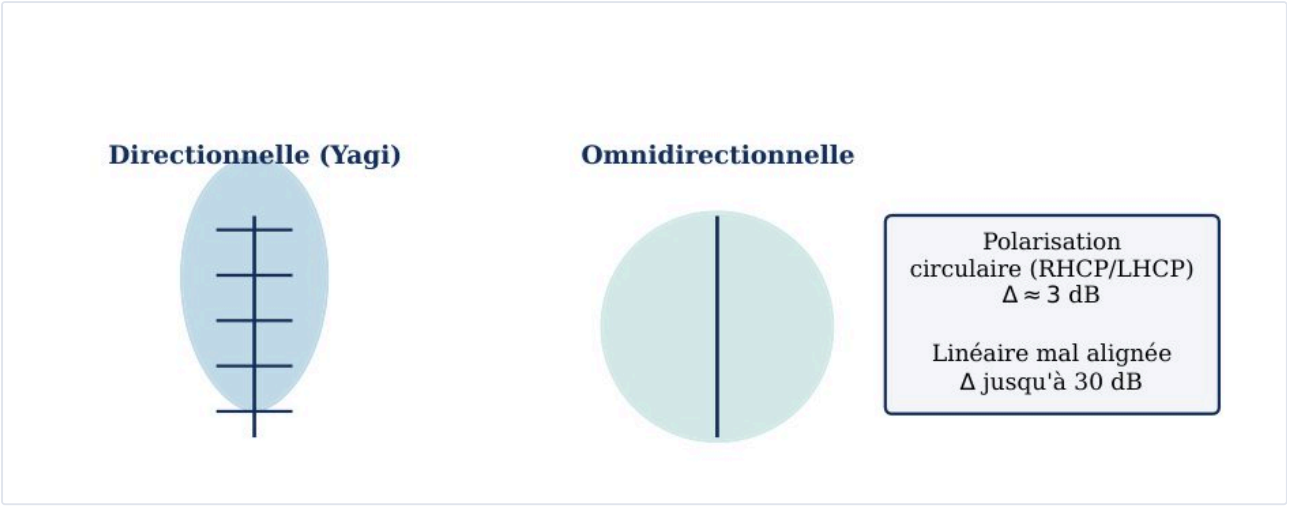


Figure 2.3 — Comparison of directional and omnidirectional antennas, with example models.

2.3 Antenna optimisation

Three levers improve reception quality:

- **Circular polarisation** — the RHCP/LHCP gap is only 3 dB against 30 dB for poorly aligned linear polarisation.
- **Gain vs bandwidth** — the higher the gain, the narrower the bandwidth. Catching very weak signals costs you bandwidth.
- **Feed line** — use low-loss coaxial cable. Since signals from space are already weak, any loss in the cable hurts.

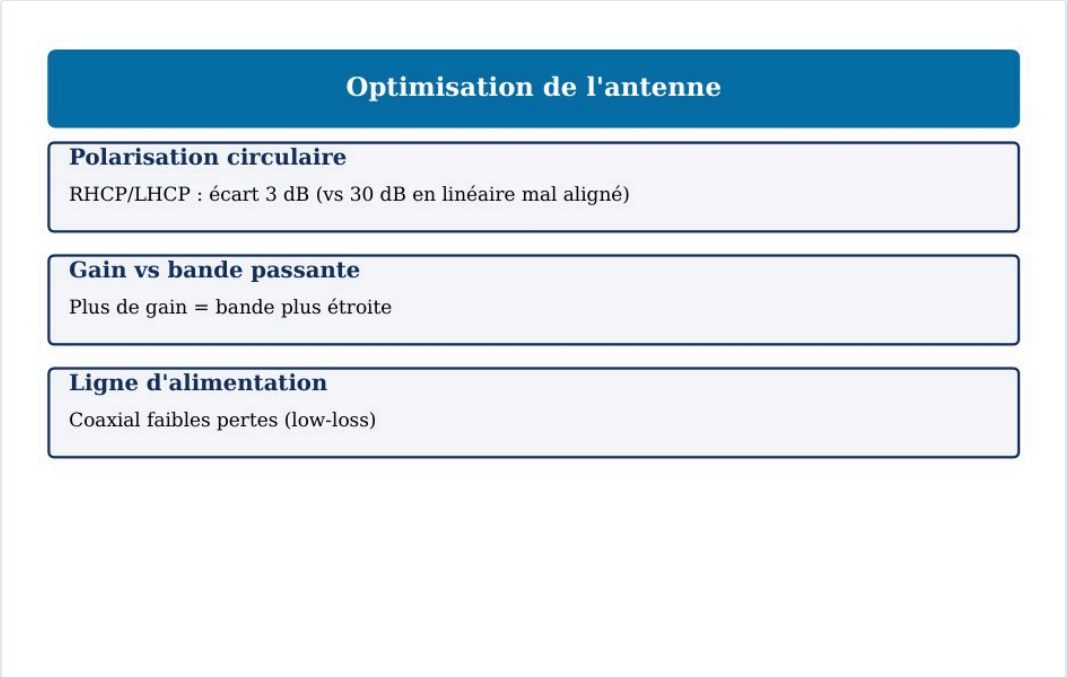


Figure 2.4 — Polarisation, the gain/bandwidth trade-off and feed-line quality.

2.4 Transponders, the Doppler effect and operating modes

The amateur-radio community describes on-board transponders as cross-band repeaters in space: they receive a signal on one frequency and re-transmit it on a different frequency and band.

Doppler effect. Just as an ambulance sounds higher-pitched as it approaches and lower as it recedes, a satellite's frequencies rise on approach then fall after it passes the zenith. The effect is stronger the higher the frequency. You therefore pre-program, in the radio, a list of frequencies to compensate for Doppler during the pass.

Operating mode. Transponders are labelled with a short identifier such as mode U/V: the first letter is the uplink band, the second the downlink band. So “U/V” means uplink in UHF and downlink in VHF. For simple listening, you care about the second letter (the downlink).

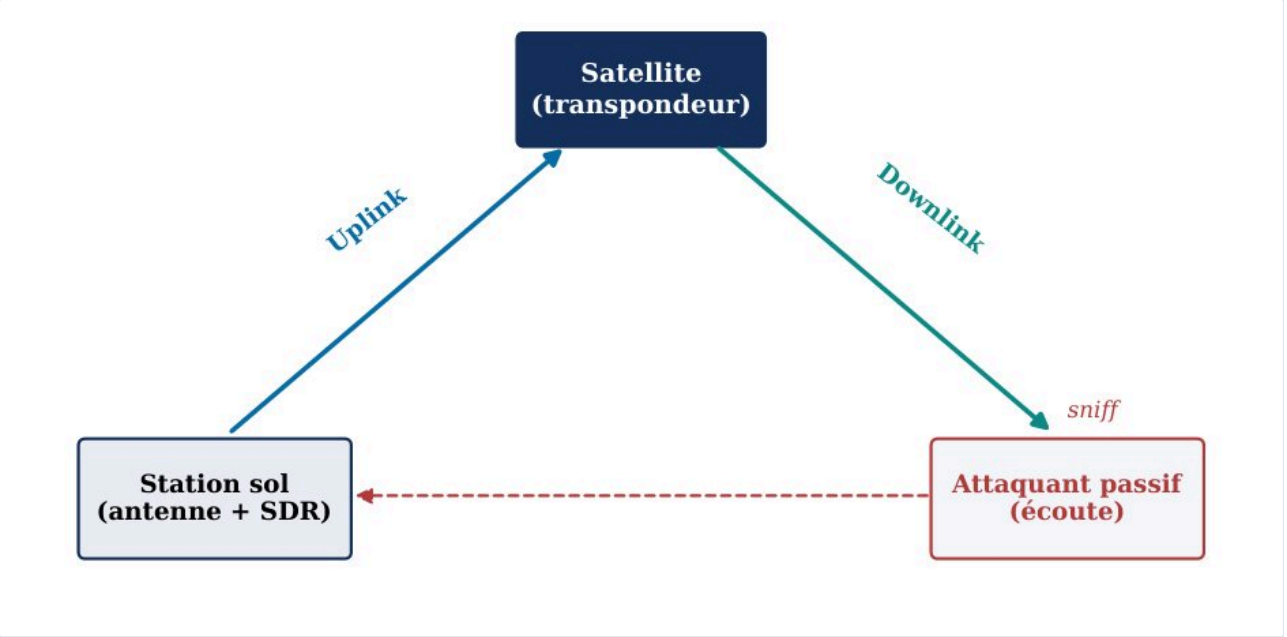


Figure 2.5 — The transponder (cross-band repeater) principle, the Doppler effect and passive eavesdropping on the downlink.

2.5 Software-Defined Radios (SDR)

SDRs are versatile, inexpensive and reliable receivers for analysing any kind of satellite communication. An SDR has a radio front-end (an RX receiver and, sometimes, a TX transmitter) with a wide tuning range, connected to a digital back-end. They handle high-rate streams, support recording and use techniques such as beamforming; they are valued for their low SWAP (Size, Weight and Power).

- **RTL-SDR** — very cheap (around $40–60), receive only, up to ~1.75 GHz, ~2.4 MHz of bandwidth. The ideal entry point, to pair with an external antenna.
- **HackRF One** — transmit and receive, 1 MHz–6 GHz, up to 20 MHz of bandwidth, more expensive.
- **Alternatives:** LimeSDR, SDRplay, SDR# (SDRSharp).

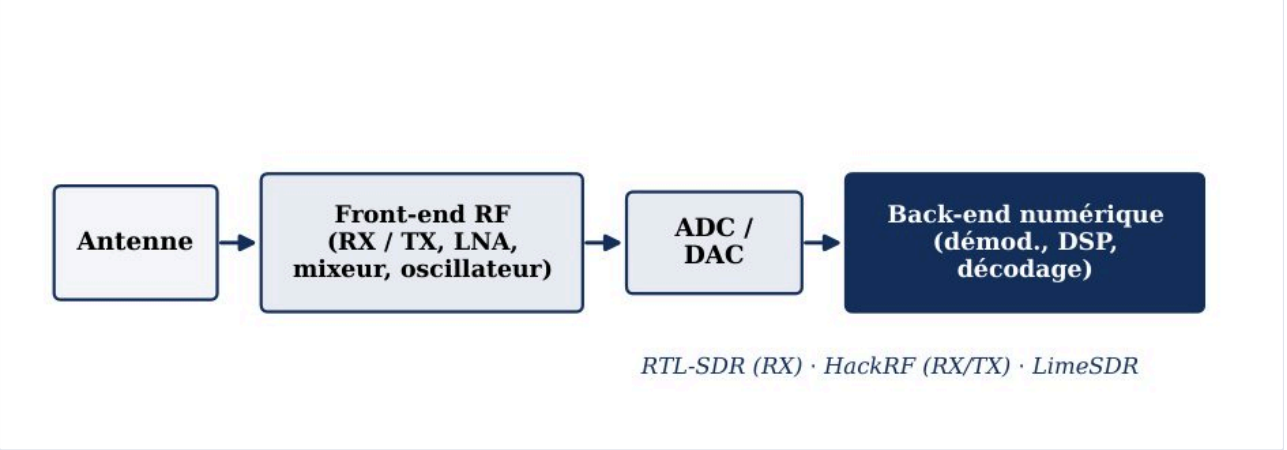


Figure 2.6 — The RX/TX architecture of an SDR and an RTL-SDR / HackRF comparison.

2.6 Beacons and the information to collect

Many satellites emit a beacon: a “bulletin board” of information on the state, health and configuration of the transponder. The main bands are 2 metres and 70 centimetres. FM satellites carry voice and packet traffic (more bandwidth, higher-quality data); SSB satellites are more spectrally efficient and carry CW, voice, SSTV or PSK31.

On the information-extraction side, a typical decoder combines a demodulator compatible with several modulation schemes, an error-correction suite and protocol decoding for telemetry frames, plus specialised image decoders and hex inspection of the signal.

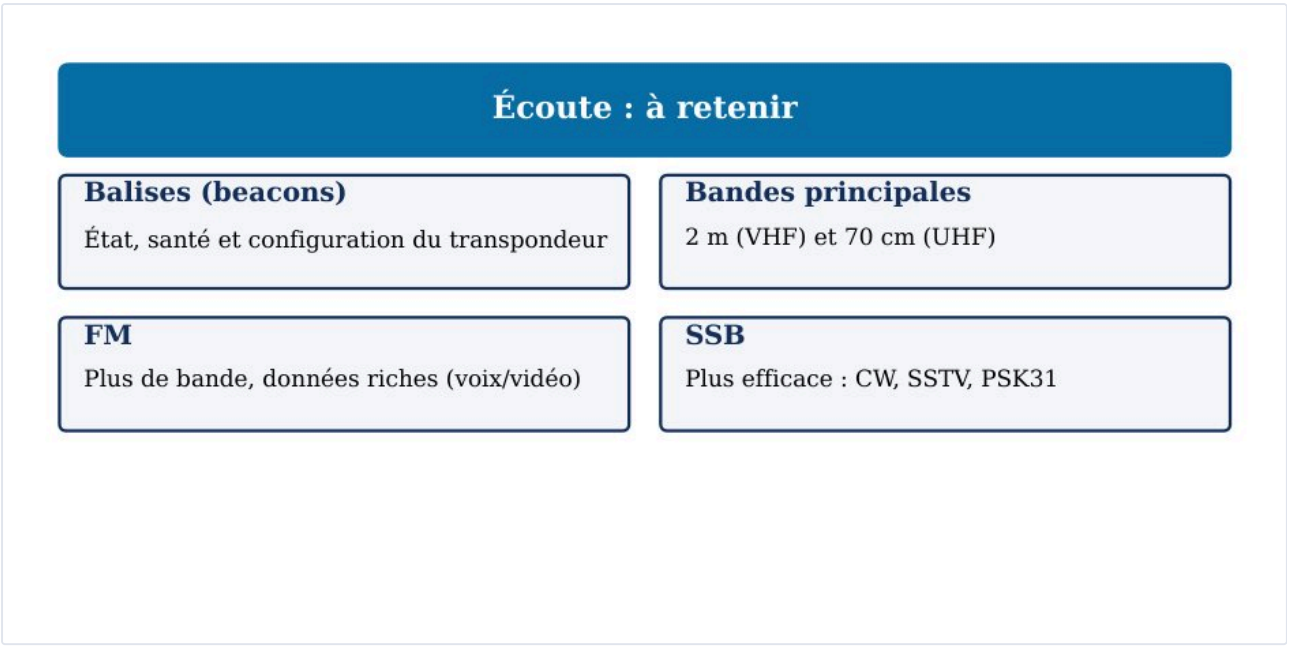


Figure 2.7 — Beacons, the main bands, and the FM / SSB distinction.

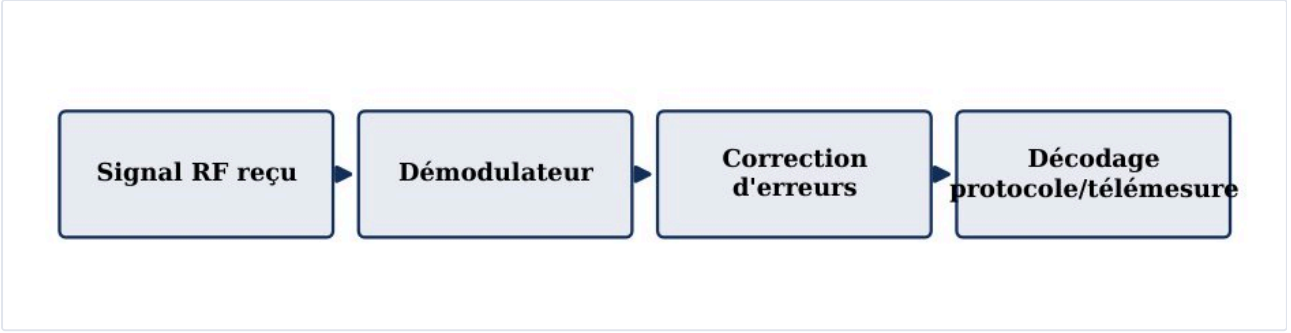


Figure 2.8 — The building blocks of a decoder: demodulation, error correction, protocol decoding.

CHAPTER 3

Reconnaissance, orbital tracking and pass prediction

The first layer of analysis is to locate the satellite and predict its passes. Without this information, you cannot aim a directional antenna or program the frequencies to compensate for Doppler.

3.1 Orbital parameters and tracking quantities

Tracking relies on Two-Line Element sets (TLE), which encode the Keplerian elements (inclination, right ascension of the ascending node, and so on). The essential quantities to track are:

- **elevation** — the angle between the observer's horizon and the satellite;
- **azimuth** — the direction of arrival measured in degrees from north, clockwise;
- the **Doppler effect**, the orbital parameters and the RSSI (Received Signal Strength Indicator);
- the **AOS** (Acquisition Of Signal), where the satellite rises above the horizon, and the **LOS** (Loss Of Signal), where it sets. You receive from AOS to LOS.

Most low-orbit satellites pass 5 to 6 times a day over a given location.

3.2 Online tools

Several free sources provide this data with no installation: N2YO [24], Heavens-Above [25] (which also guides antenna pointing), OrbTrack, and AMSAT's pass-prediction tool [23]. You check these sources several times a day before a listening session. For more permanent use, applications such as JSatTrak or SatHunter install locally and offer ground-track and radiation-pattern visualisations.

3.3 gpredict: real-time prediction and tracking

gpredict [6] is a desktop application for real-time satellite tracking and orbit prediction, based on the SGP4/SDP4 propagation algorithms and NORAD TLEs. Unlike online services, it offers richer information: display on graphs and maps, prediction of many successive passes, and full modularity.

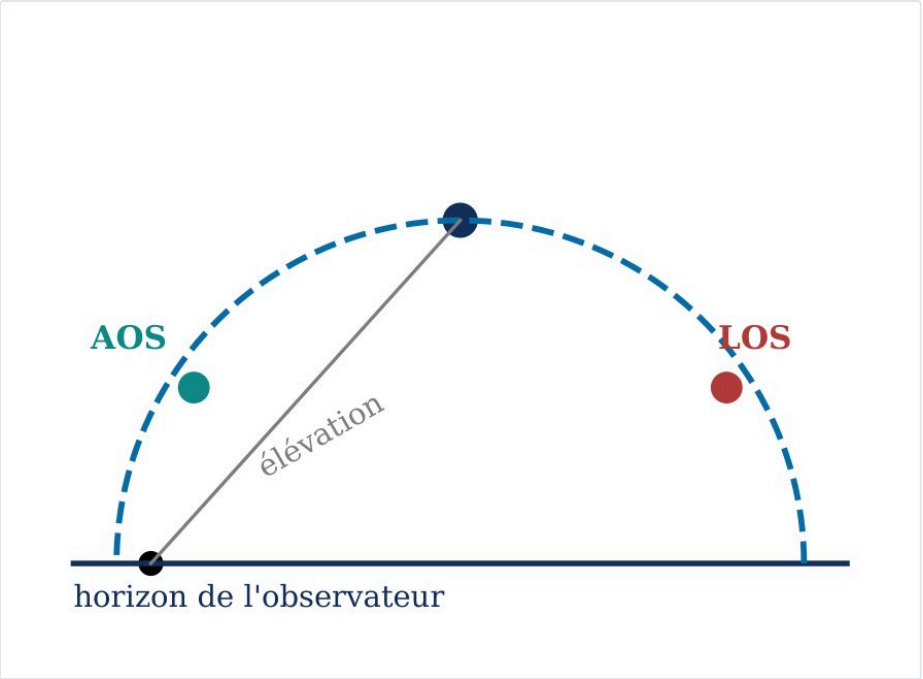


Figure 3.1 — Tracking quantities: azimuth, elevation, Doppler, AOS/LOS.

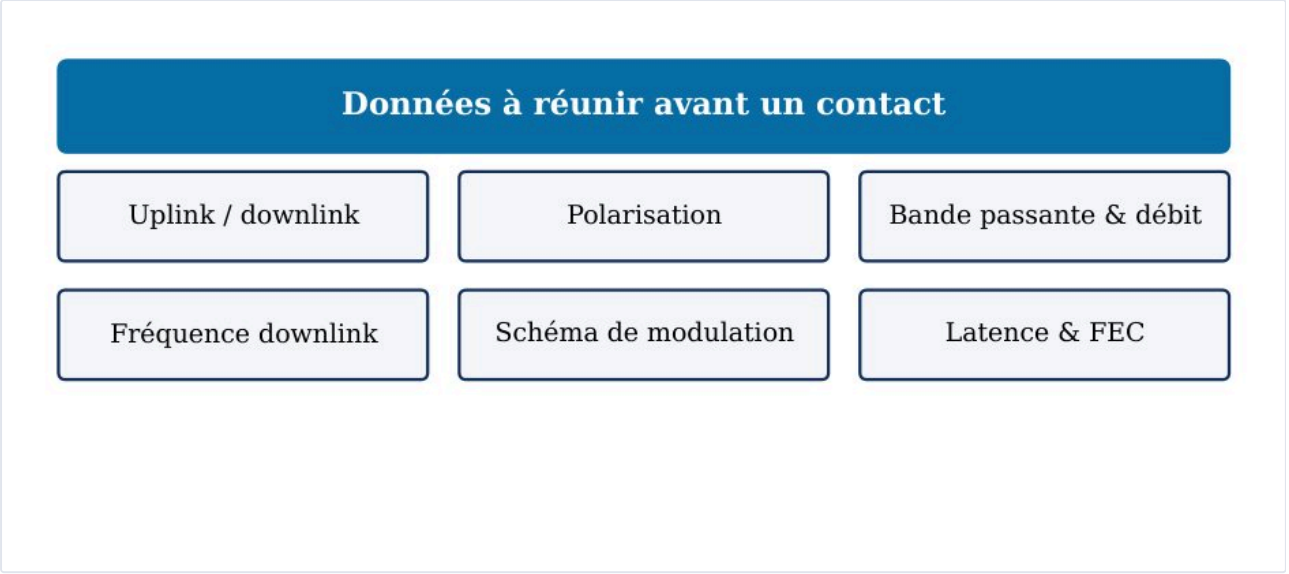
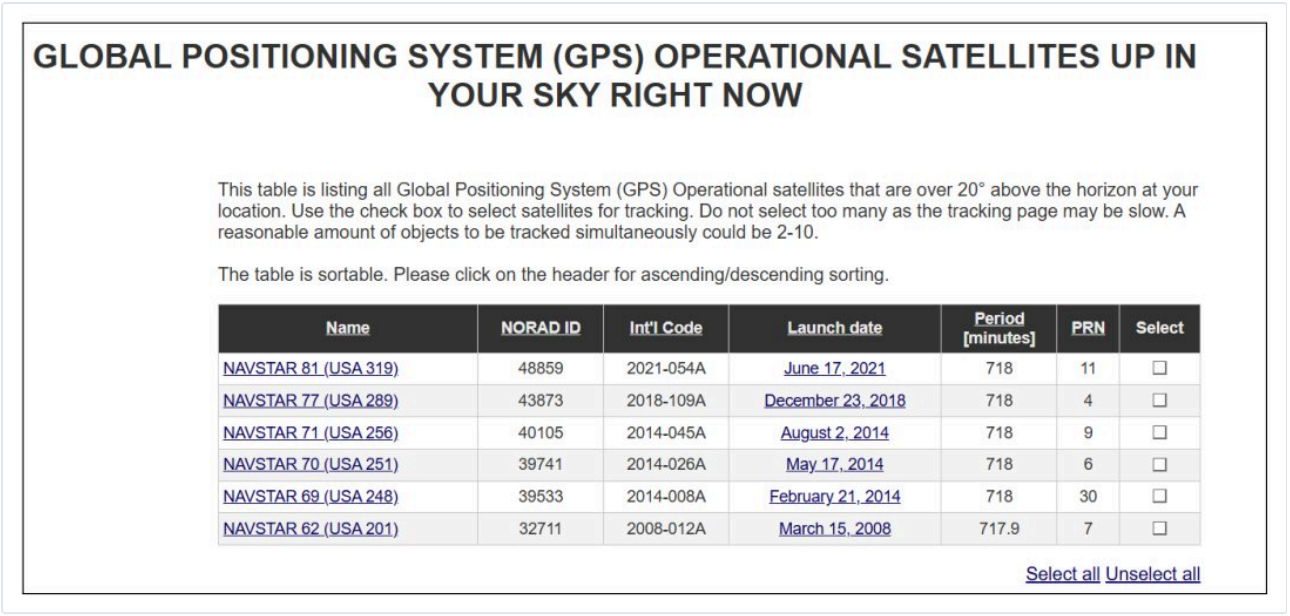


Figure 3.2 — Data to gather before a contact: uplink/downlink, polarisation, bandwidth, rate, modulation, latency, FEC.



GLOBAL POSITIONING SYSTEM (GPS) OPERATIONAL SATELLITES UP IN YOUR SKY RIGHT NOW

This table is listing all Global Positioning System (GPS) Operational satellites that are over 20° above the horizon at your location. Use the check box to select satellites for tracking. Do not select too many as the tracking page may be slow. A reasonable amount of objects to be tracked simultaneously could be 2-10.

The table is sortable. Please click on the header for ascending/descending sorting.

Name	NORAD ID	Int'l Code	Launch date	Period [minutes]	PRN	Select
NAVSTAR 81 (USA 319)	48859	2021-054A	June 17, 2021	718	11	☐
NAVSTAR 77 (USA 289)	43873	2018-109A	December 23, 2018	718	4	☐
NAVSTAR 71 (USA 256)	40105	2014-045A	August 2, 2014	718	9	☐
NAVSTAR 70 (USA 251)	39741	2014-026A	May 17, 2014	718	6	☐
NAVSTAR 69 (USA 248)	39533	2014-008A	February 21, 2014	718	30	☐
NAVSTAR 62 (USA 201)	32711	2008-012A	March 15, 2008	717.9	7	☐

Select all Unselect all

Figure 3.3 — A web service for pass prediction and TLE sets (N2YO).

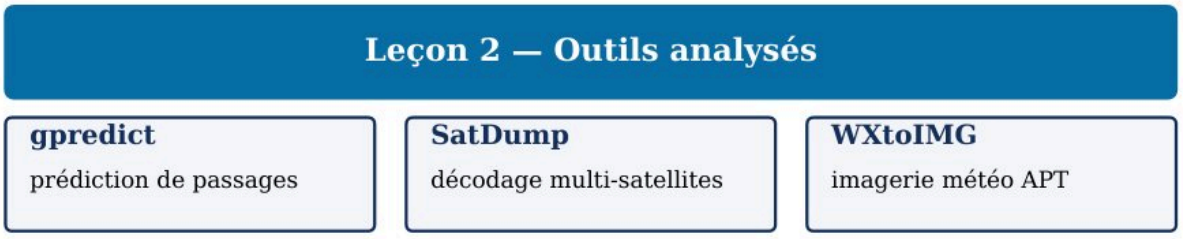


Figure 3.4 — The three open-source tools studied: gpredict, SatDump and WXtoIMG.

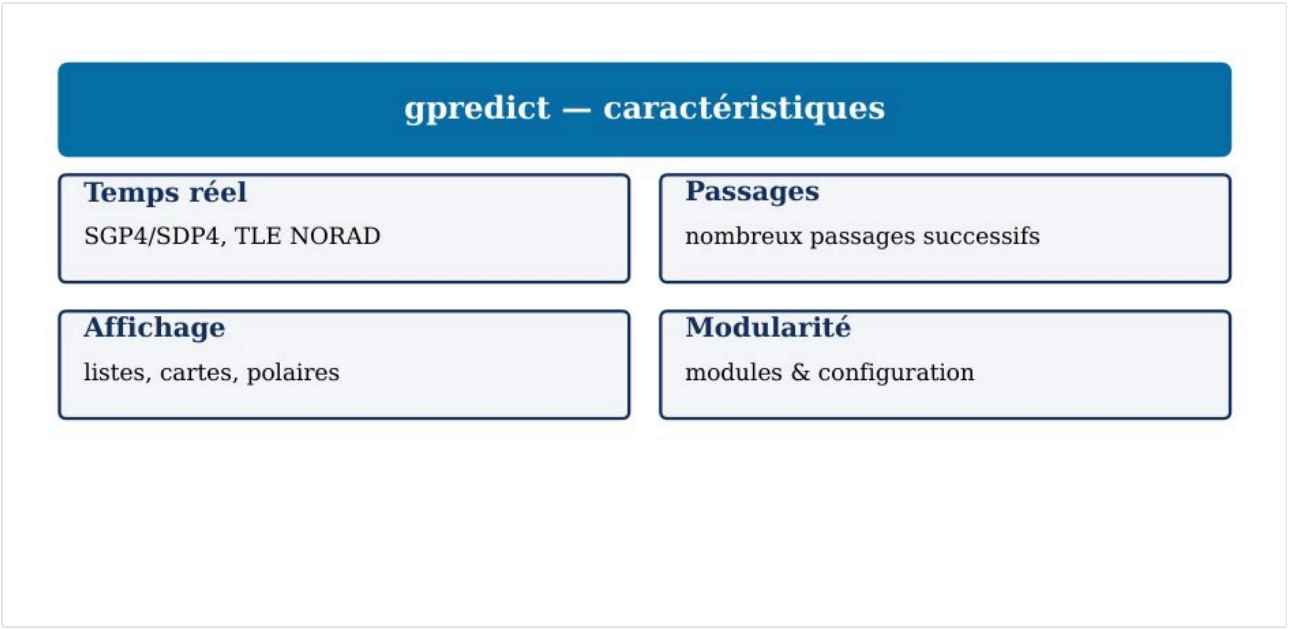


Figure 3.5 — gpredict's main features: real time, multiple passes, modularity.

Installation. On Windows, download the archive from SourceForge; on macOS, via MacPorts. The first step is to update the data: Edit → Update TLE data from network, then Update transponder data.

Create a module and read the data. Create a module (File → New module), for example “NOAA”, adding NOAA 15, 18 and 19. Selecting a satellite shows azimuth, elevation, range rate, SSP lock, ground footprint, Doppler at 1000 m and orbital phase. A right-click → Future passes lists upcoming passes and their duration.

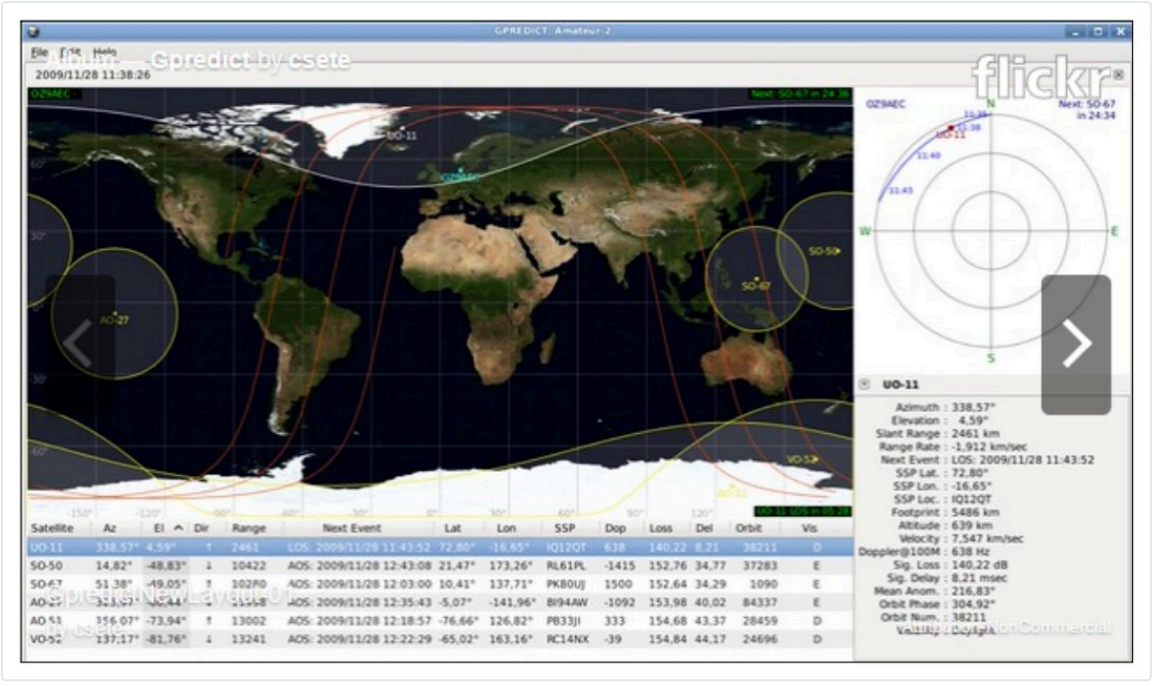


Figure 3.6 — The gpredict download/installation page, by operating system.

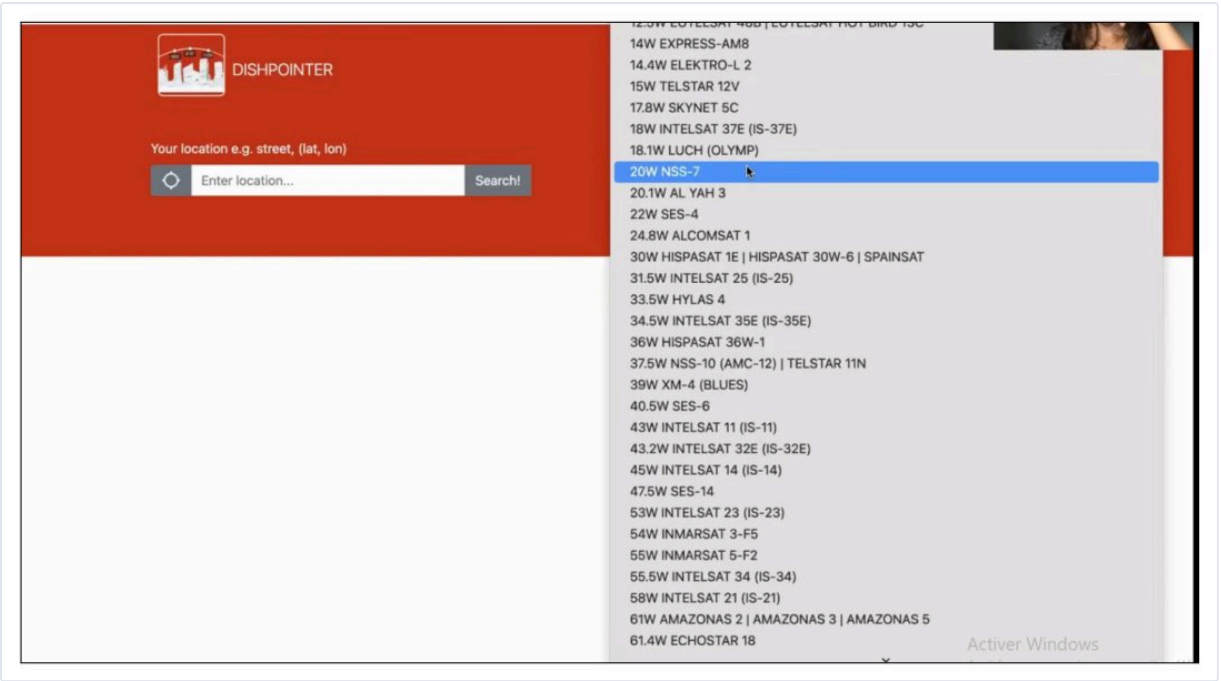


Figure 3.7 — gpredict installation/configuration (SourceForge distribution, transponder data).

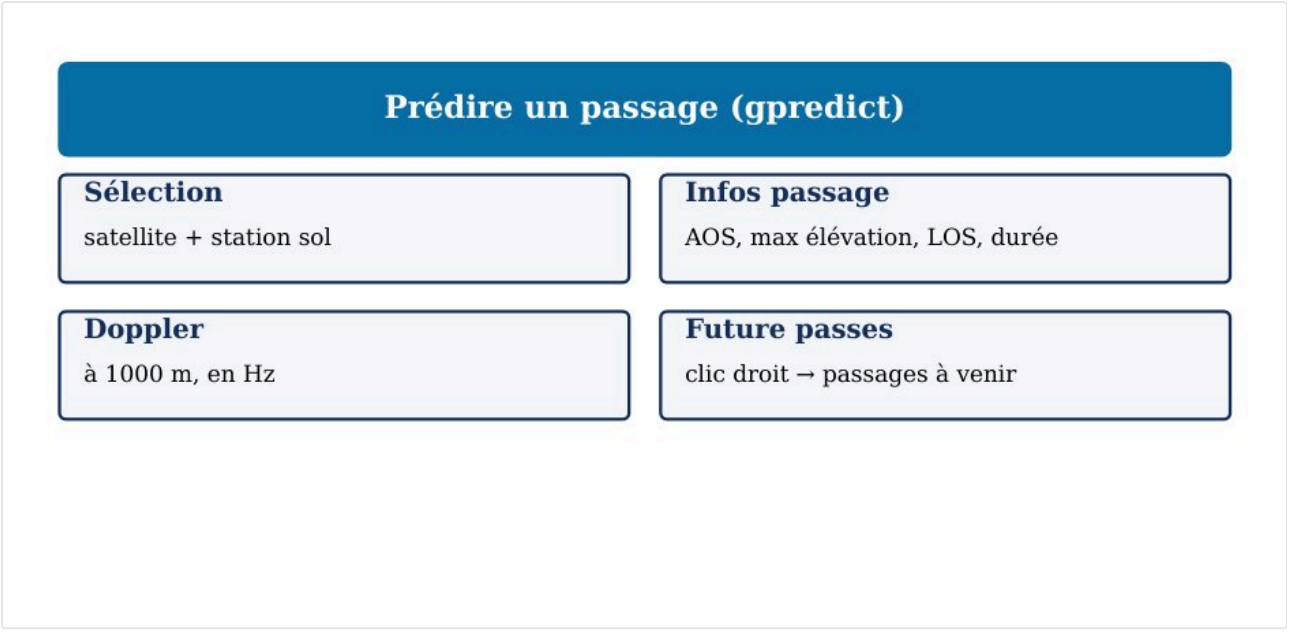


Figure 3.8 — A pass-prediction view (polar plot) in gpredict.

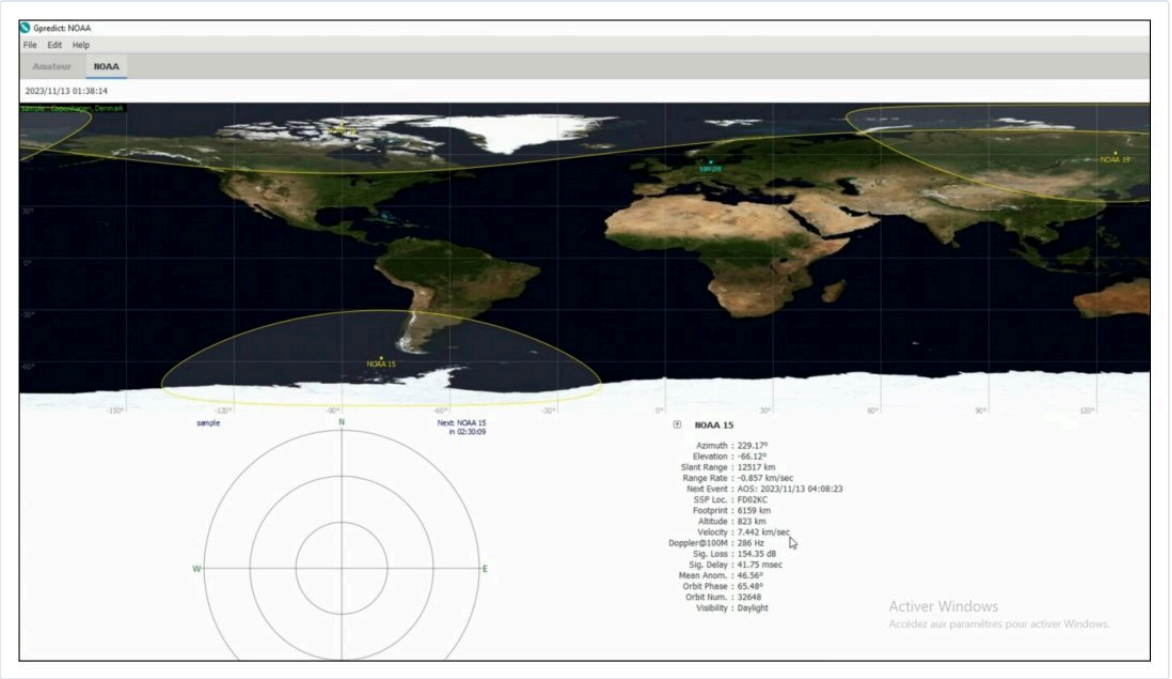


Figure 3.9 — gpredict: the satellite's ground track projected on the world map.

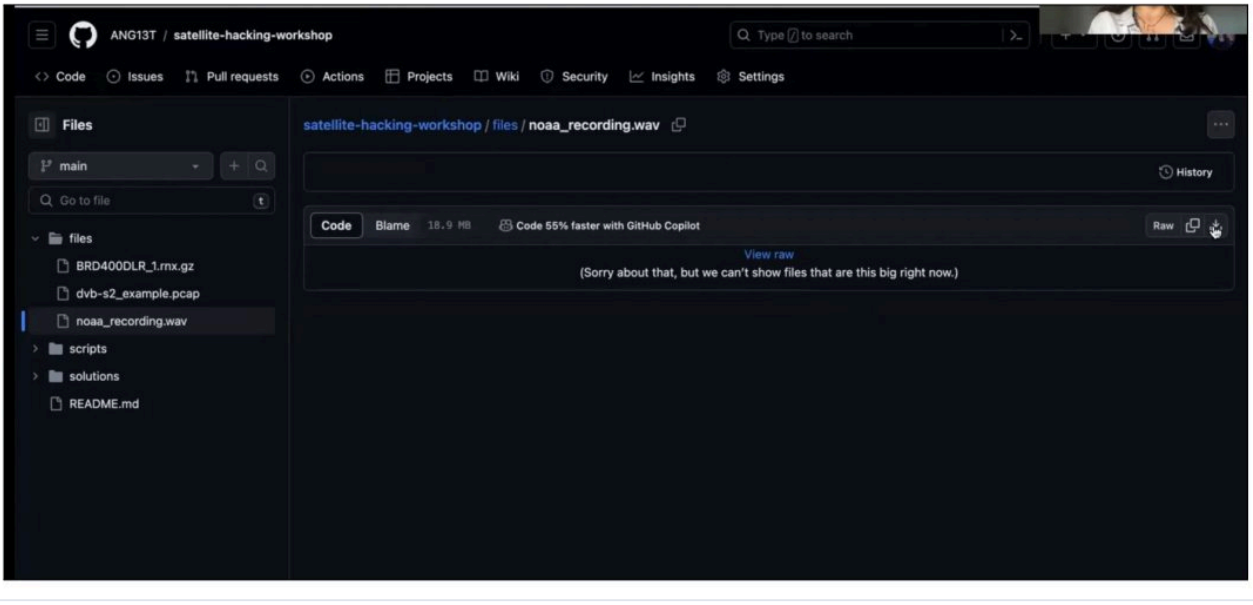


Figure 3.10 — gpredict: a polar plot of a pass (elevation and azimuth over time).

3.4 DishPointer: aiming a dish

For a parabolic antenna, the DishPointer site [7] provides precise settings: elevation angle, azimuth (true and magnetic) and LNB skew. You select the target satellite and enter the latitude/longitude; a line of sight shows the orientation, with an “obstacles” option to spot anything that could attenuate the signal.

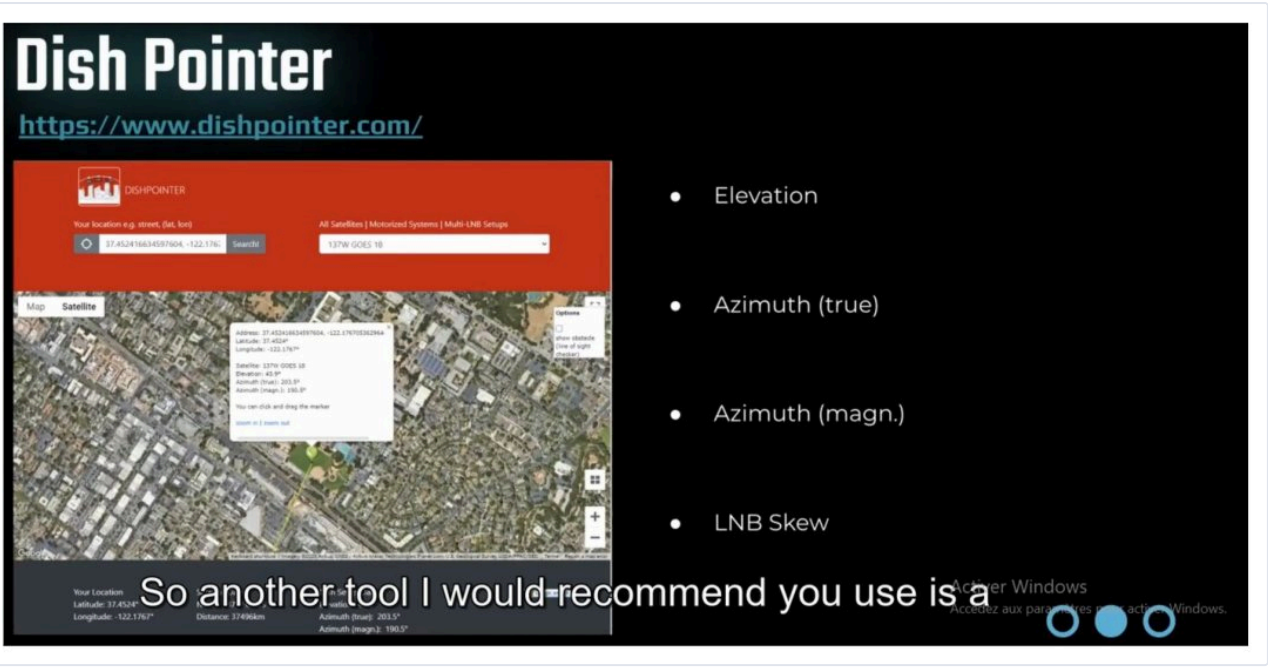


Figure 3.11 — A dish-alignment calculator (elevation, azimuth, LNB skew).

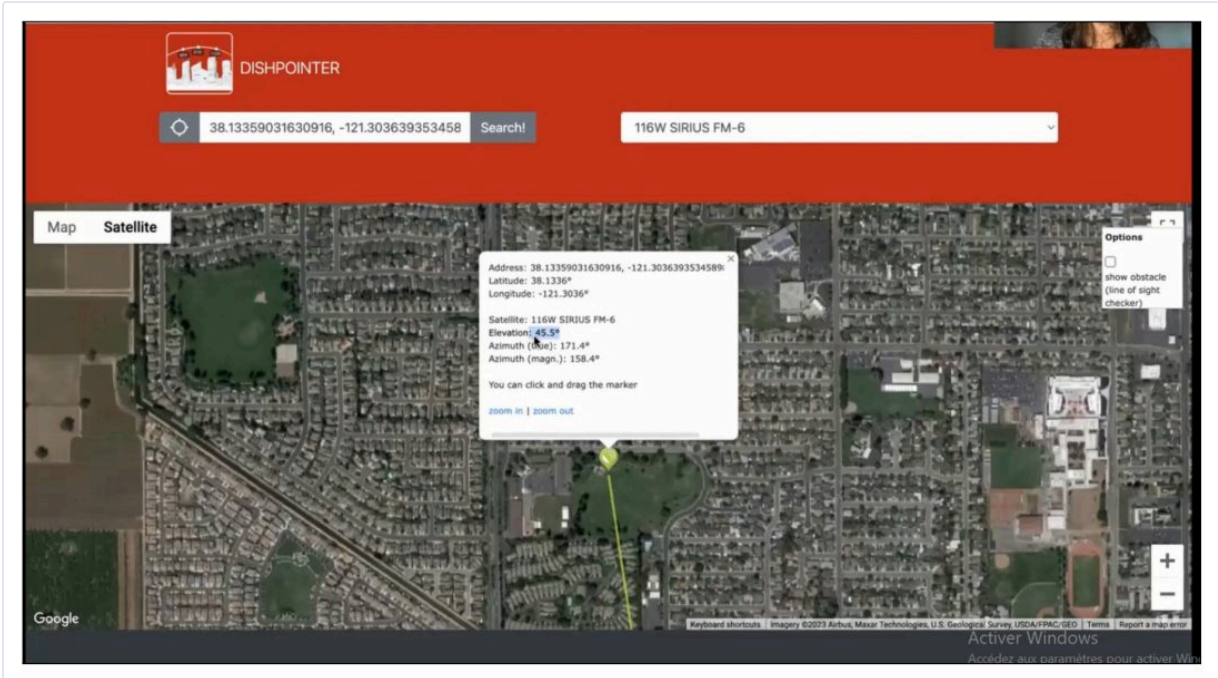


Figure 3.12 — DishPointer: the satellite line of sight drawn on a map view.



Figure 3.13 — Antenna options (QFH, mesh, dish) and elevation/azimuth/skew setting.

CHAPTER 4

Decoders and signal acquisition

4.1 Anatomy of a satellite decoder

A decoder turns a received radio signal into usable data. A good decoder brings together three functions [4]:

1. **Demodulation** — extracting the baseband signal from the modulated carrier;
2. **Forward error correction (FEC)** — detecting and correcting transmission errors with error-correcting codes;
3. **Protocol decoding** — recovering the data formatted according to the satellite's protocol.

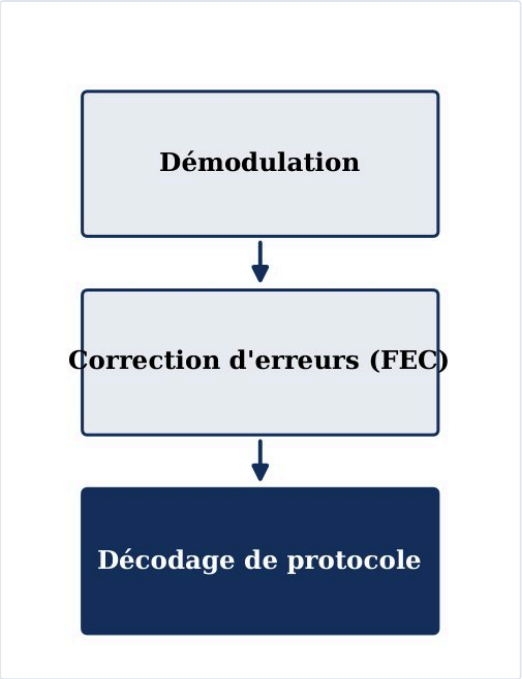


Figure 4.1 — The functions expected of a decoder: demodulation, error correction, protocol decoding.

4.2 Tool landscape

Several families of tools can be distinguished, covered in detail in the following chapters:

- **Multi-format decoders:** SatNOGS, GNU Radio and its gr-satellites module, the SDR# suite, Universal Radio Hacker (URH).
- **Packet and frame analysis:** hexdump for hex inspection, Wireshark, the Open Satellite Project.
- **Reverse engineering:** SatDump (offline processing), Ghidra for firmware and binaries.

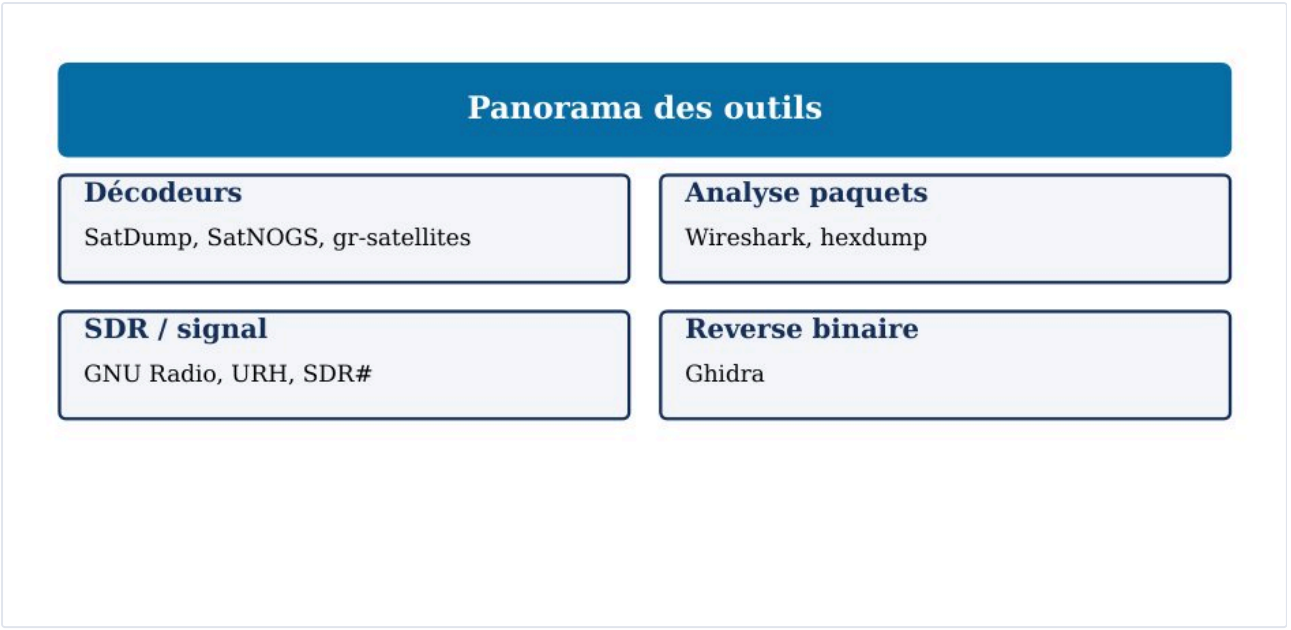


Figure 4.2 — Categories of tools for analysing satellite communications.

4.3 Recording a transmission

To capture a signal, you use an SDR program such as GQRX. You select the correct frequency for the satellite (e.g. NOAA 15 downlink at 137.62 MHz), choose WFM mono mode, a wide filter width, a suitable reference level and dB range, and enable peak detection. The LNA gain is typically set around 20 dB for best results.

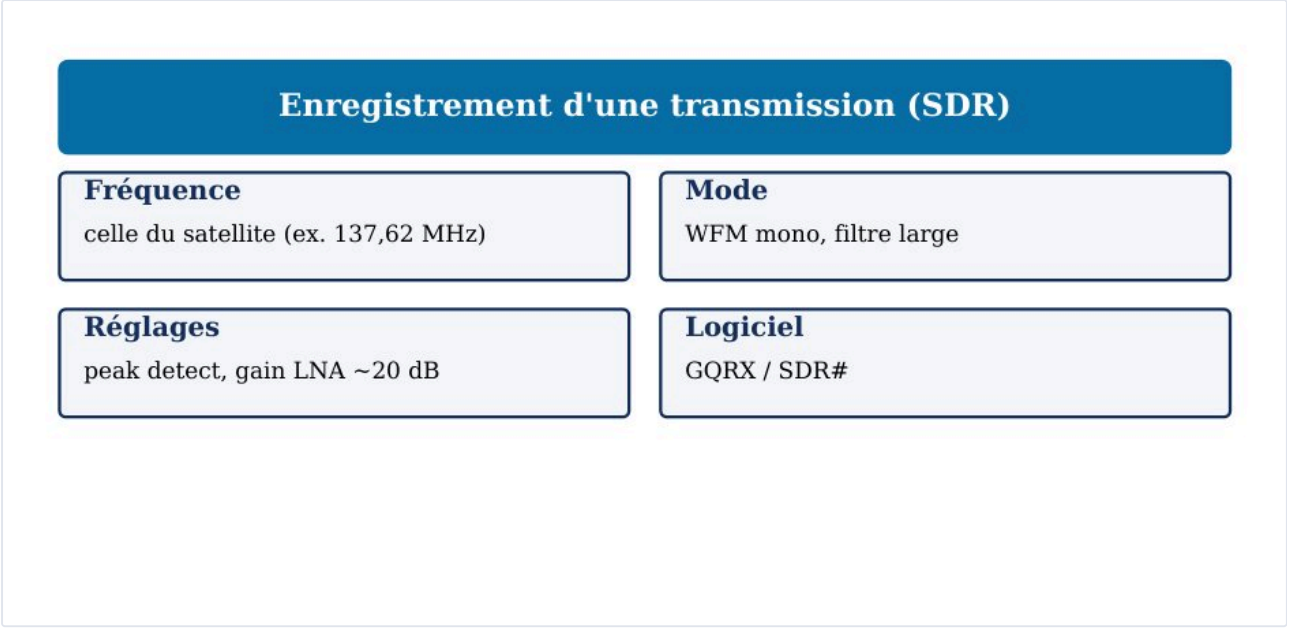


Figure 4.3 — Configuring an SDR receiver (waterfall) for recording.

CHAPTER 5

Weather imagery: the APT protocol and its decoding

5.1 The APT protocol

APT (Automatic Picture Transmission) is used by weather satellites, notably the NOAA fleet, to transmit weather and remote-sensing images to ground stations. The structure is a broadcast with two channels: a telemetry channel (A and B frames) and a synchronisation channel.

Characteristics. The image is sent as horizontal scan lines, at 2 lines per second — i.e. 4160 baud [8]. Documented downlink frequencies: NOAA 15 at 137.62 MHz, NOAA 18 at 137.9125 MHz, NOAA 19 at 137.10 MHz (VHF band).

Current status. This mode is now historic: the last satellite transmitting in APT, NOAA 15, was decommissioned on 19 August 2025 [8]. APT nonetheless keeps great teaching value for understanding the decoding of analogue images by SDR.

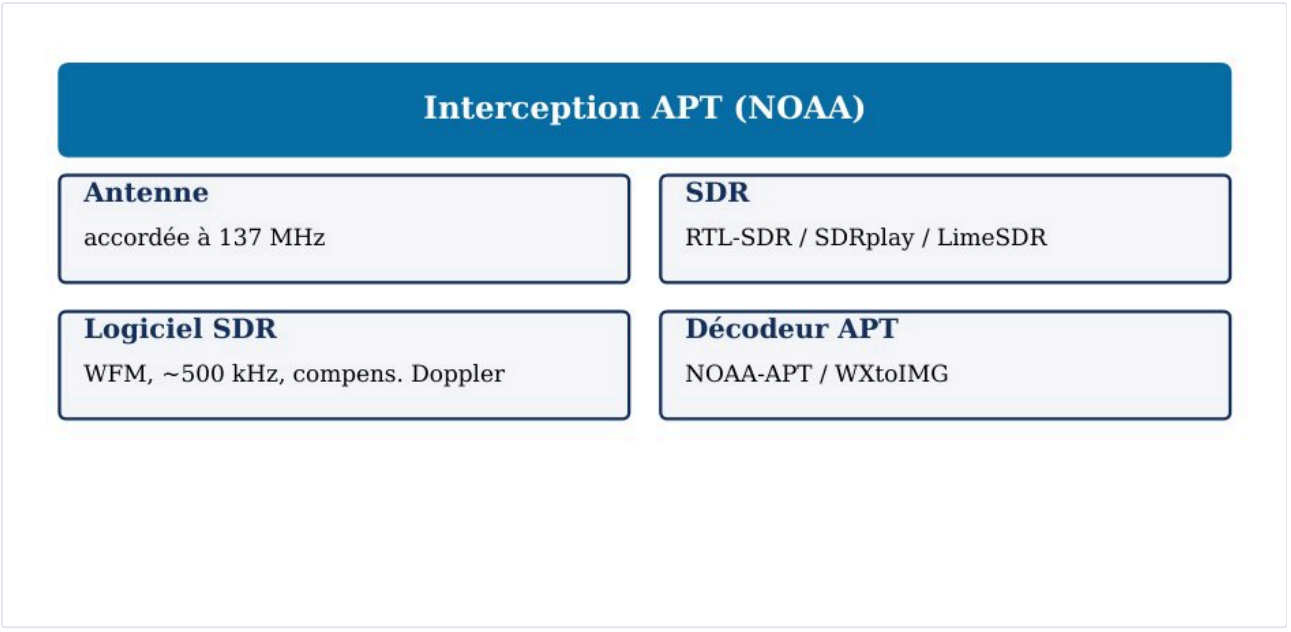


Figure 5.1 — The APT interception principle: a 137 MHz antenna, an SDR and decoding software.

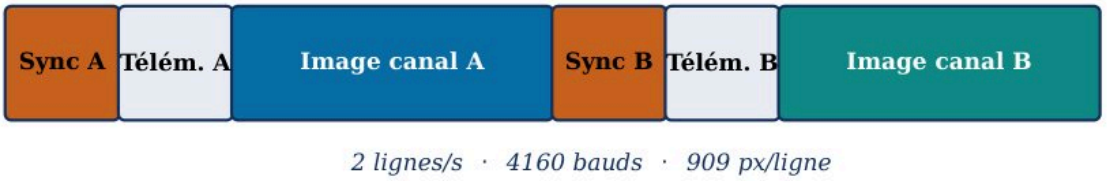


Figure 5.2 — The structure of an APT transmission (A/B channels and sync, scan lines).

5.2 WXtoIMG and online decoders

WXtoIMG [9] is a long-standing decoder of weather-satellite signals: automated recording and decoding, map projections, map overlays, temperature/sensor calibration and command-line control. The software was abandoned by its author in 2018; a community rebuild, WXtoImg Restored, redistributes its binaries [9]. In practice, you can use an online APT decoder: load the raw file of a NOAA recording, apply the absolute-value option then decode, to obtain the images (A and B channels, sync adjustment).

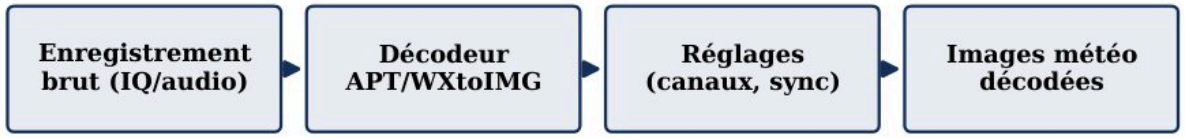


Figure 5.3 — A weather image reconstructed from a NOAA recording.

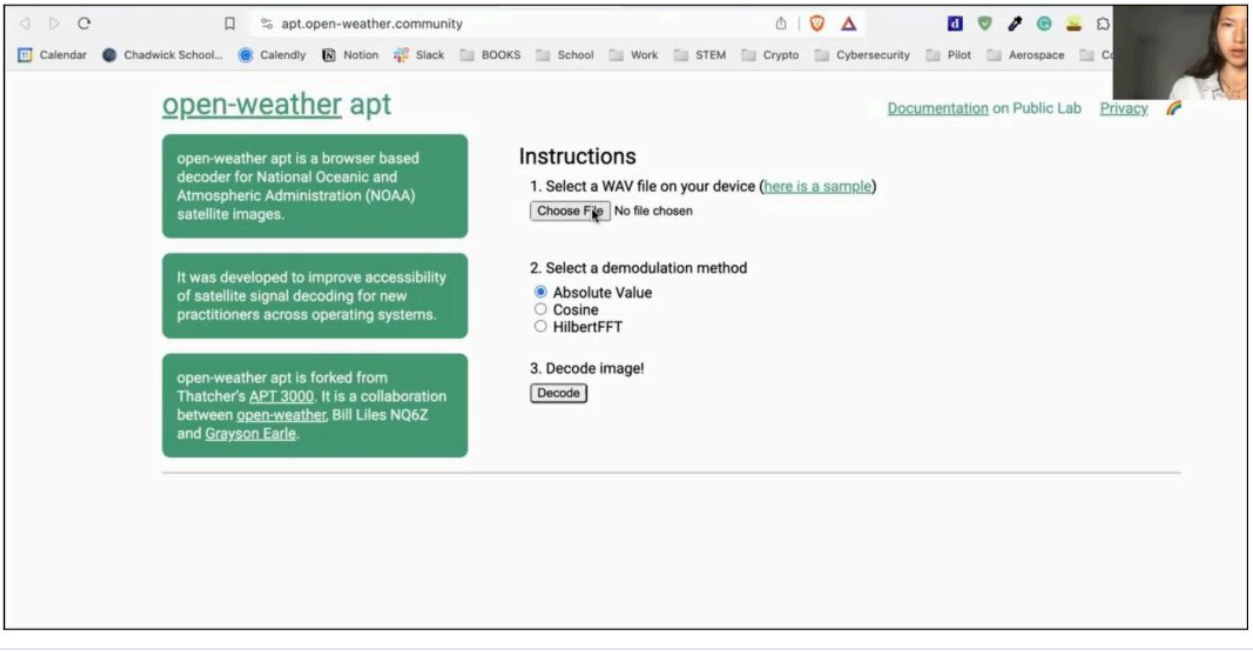


Figure 5.4 — An online APT decoder used as an alternative to WXtoIMG.

5.3 SatDump: a satellite-data processing suite

SatDump [4] is a robust, generic tool for processing satellite data via SDR. Auto-detected on connection (RTL-SDR, LimeSDR, HackRF…), it decodes and processes the transmissions of more than 90 satellites and probes. It handles live links such as APT, LRPT, HRPT, LRIT/HRIT, decodes weather-constellation imagery and overlays images on a map with instrument layers.

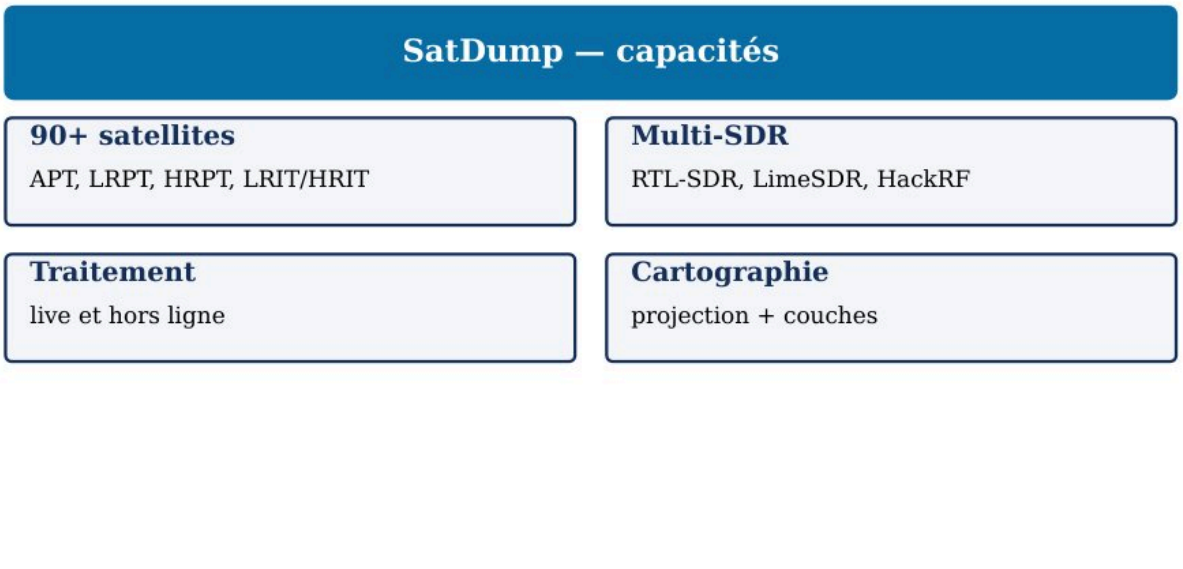


Figure 5.5 — An overview of SatDump and its capabilities (90+ satellites, multi-SDR, offline).

Lab: offline APT decoding. The typical procedure: select Offline processing, find the “NOAA APT” pipeline, choose the raw input file and an output folder, set the sample rate (e.g. 11025 KSPS depending on the recording), then Start. The result is viewed in the Viewer; processing (equalisation, RGB composites, projections) improves legibility.

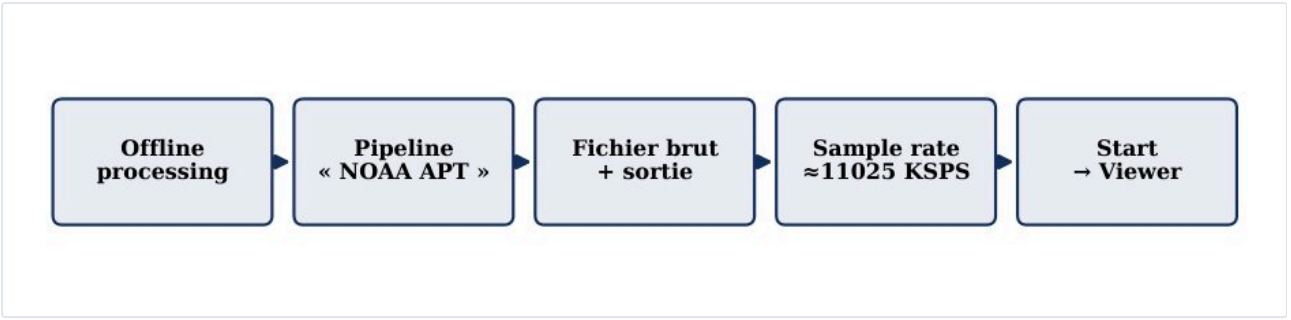


Figure 5.6 — Setting up APT processing in SatDump.

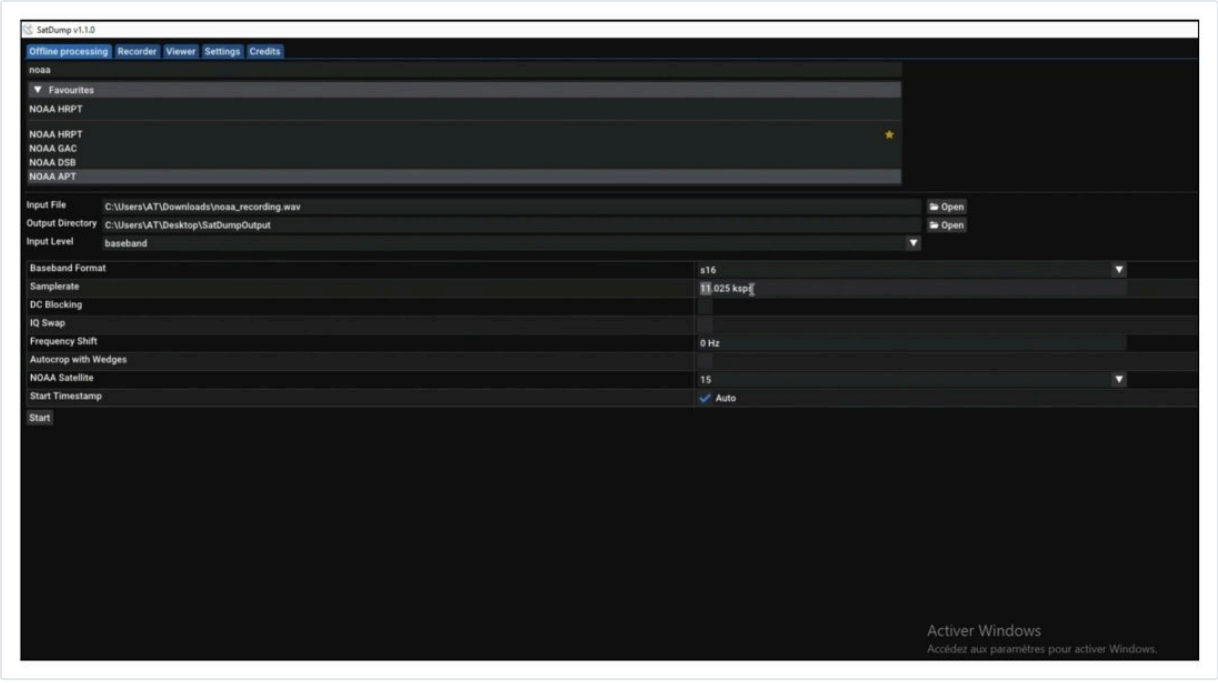


Figure 5.7 — SatDump: the processing interface for a satellite transmission.

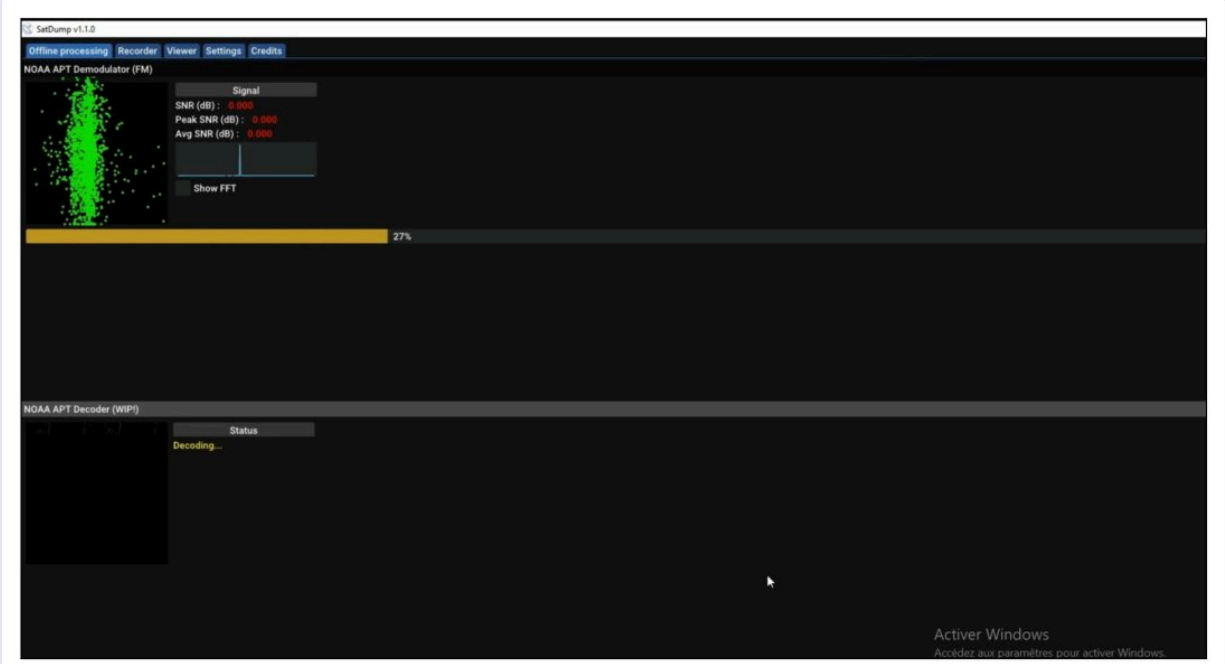


Figure 5.8 — SatDump: visualising the signal during processing (waterfall).

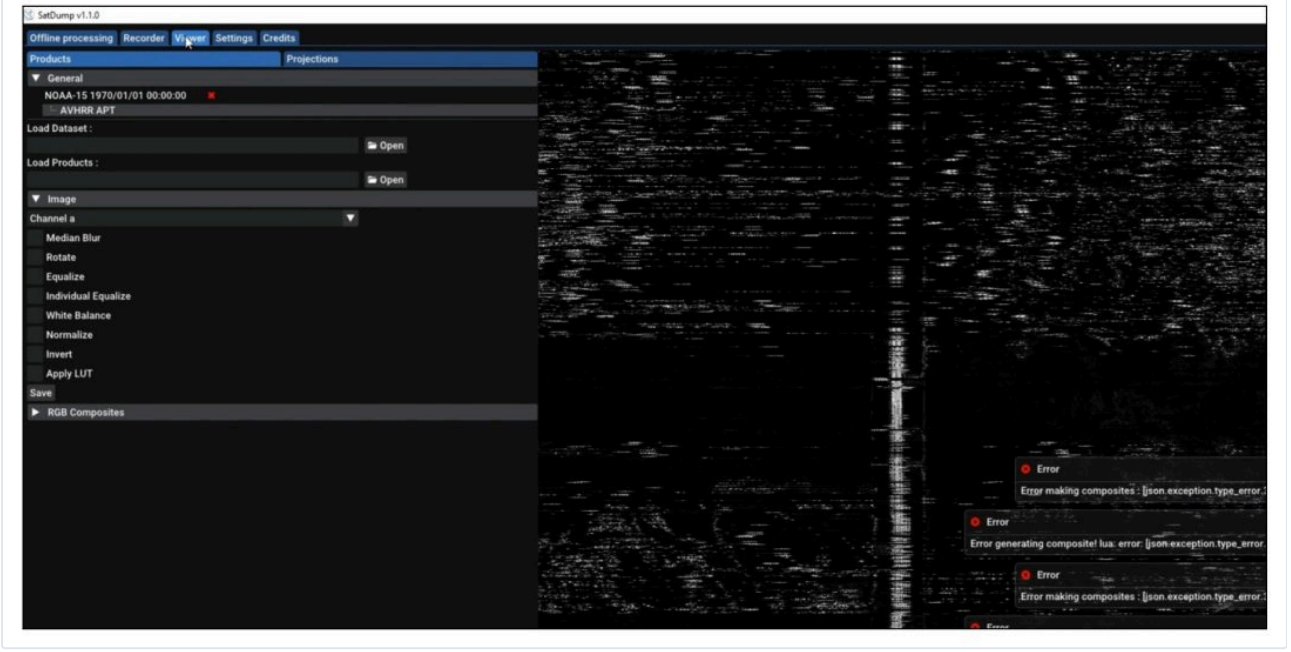


Figure 5.9 — SatDump: the APT data processing/decoding step.

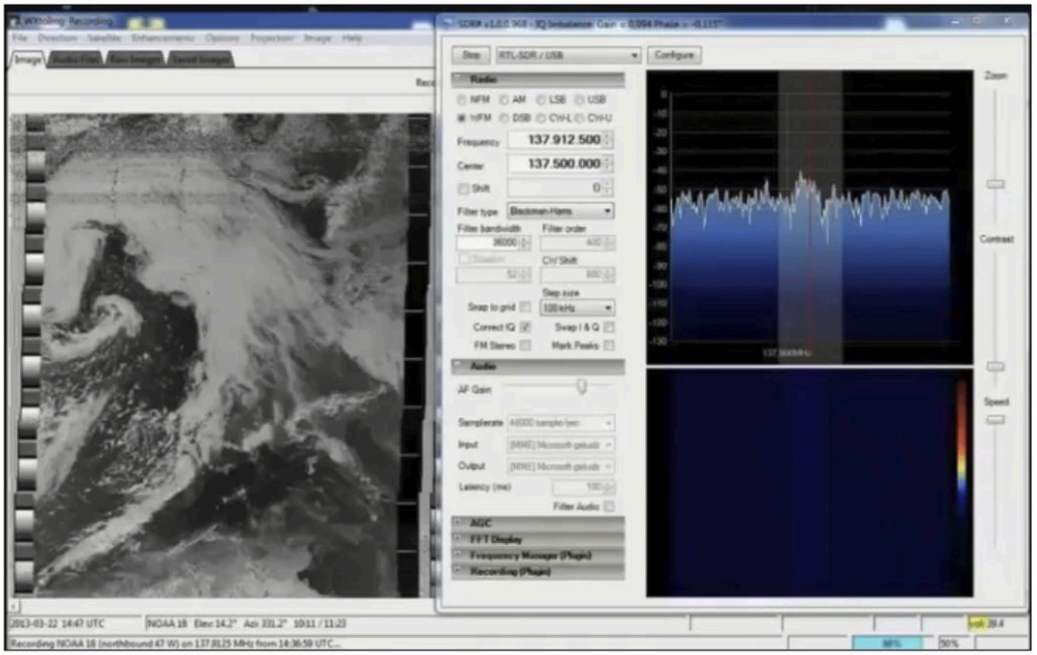


Figure 5.10 — Decoded NOAA weather images output by SatDump.

CHAPTER 6

DVB: digital video broadcasting and interception

DVB (Digital Video Broadcasting) is a very widespread audio/video transmission protocol. This chapter presents its principles, its frame structure, satellite Internet-access architectures, then the interception toolchain (GNU Radio, Wireshark) and the associated network threats.



Figure 6.1 — The DVB part's plan: principles, DVB-S2 extraction via GNU Radio, Wireshark dissection.

6.1 DVB-S / DVB-S2 principles

DVB carries audio and video over data links. The satellite variant is DVB-S (distinct from terrestrial DVB-T). The modern DVB-S2 version is about 30% more efficient thanks to its formatting. DVB uses a Coded OFDM modulation [15]. Each DVB satellite carries several transponders operating on different frequencies, most often in the C and Ku bands (sometimes Ka).

6.2 The MPEG-2 transport stream

The DVB-S transport stream (TS) is a digital format carrying audio, video and data. A multiplexer combines several sources (films, audio, data, sport…) into a single continuous stream of fixed-size packets.

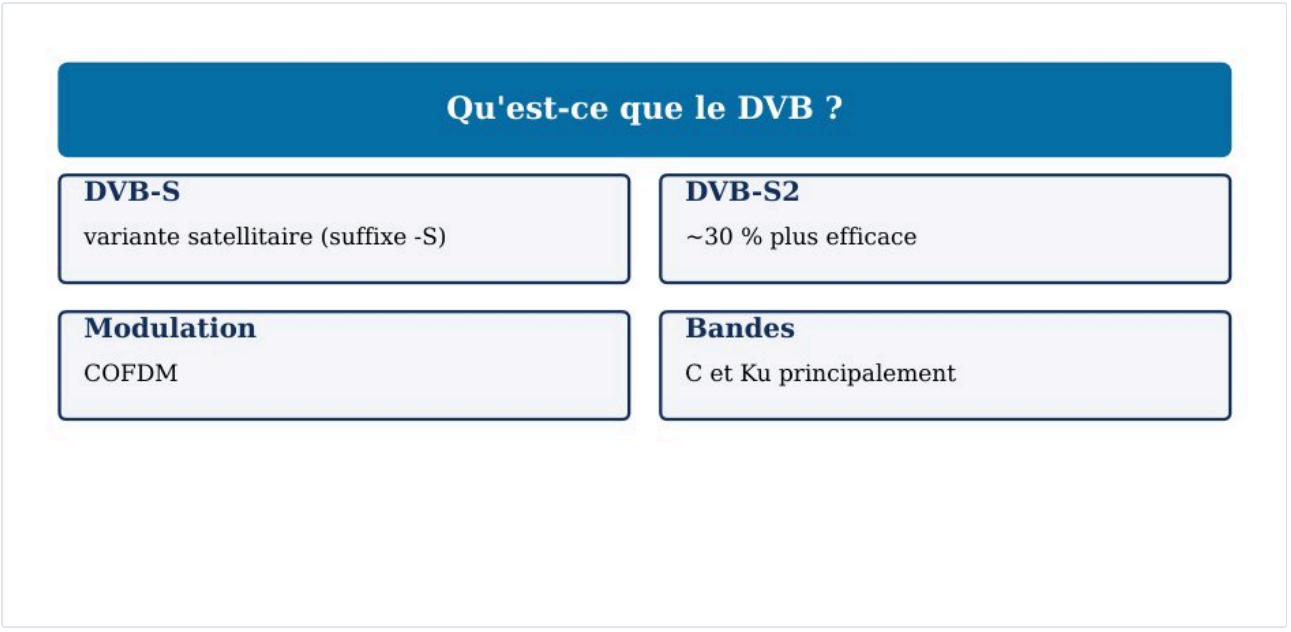


Figure 6.2 — Definition of DVB and its satellite variant DVB-S/-S2.

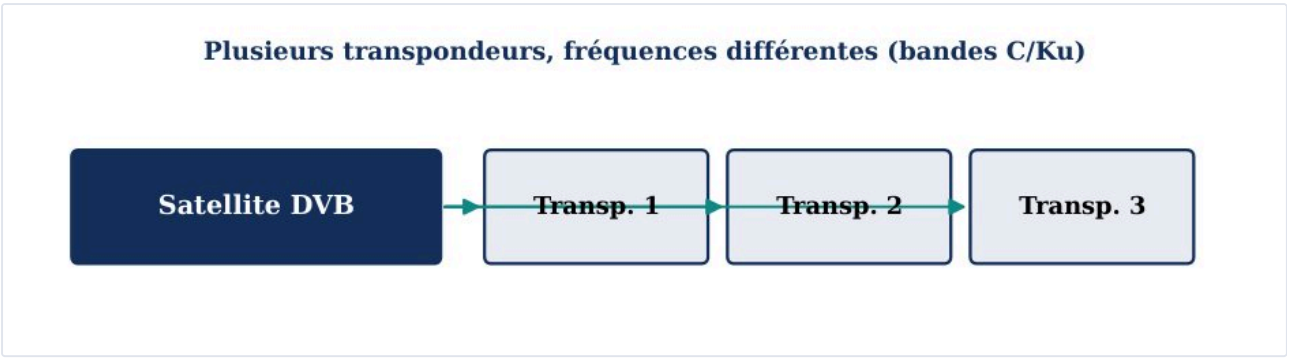


Figure 6.3 — A DVB satellite's multiple transponders and the bands used (C, Ku).

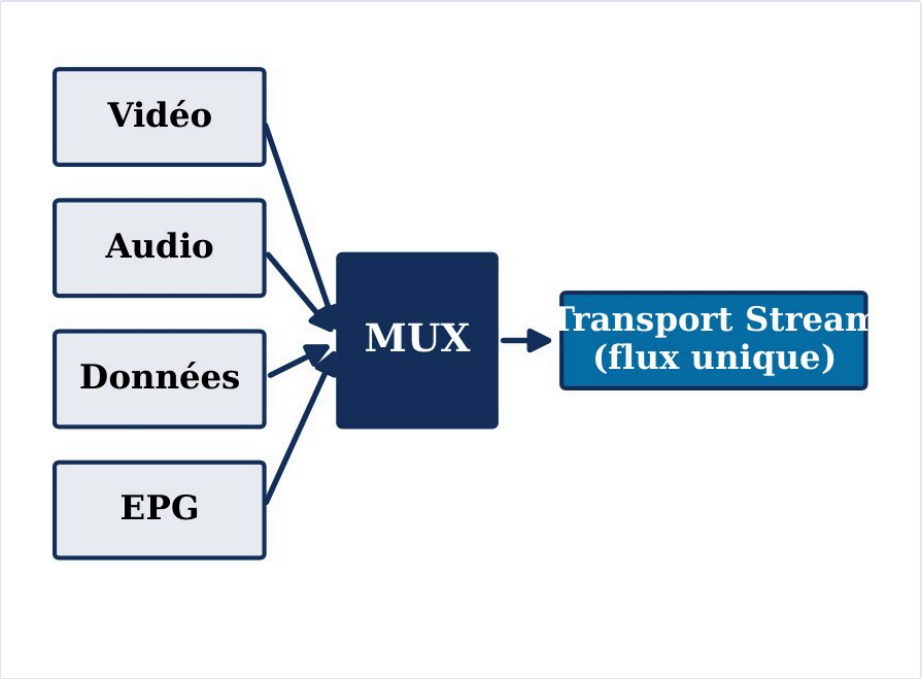


Figure 6.4 — Multiplexing heterogeneous sources into a single transport stream.

Anatomy of an MPEG-2 TS packet. Each packet is 188 bytes and starts with a sync byte of value 0x47 [14]. It is followed by the Transport Error Indicator (TEI), the Payload Unit Start Indicator (PUSI), the Transport Priority, the 13-bit PID (Packet Identifier) that identifies the elementary stream, the Transport Scrambling Control (2 bits, key-based scrambling), the Adaptation Field Control (2 bits) and the Continuity Counter (the payload's sequence number).

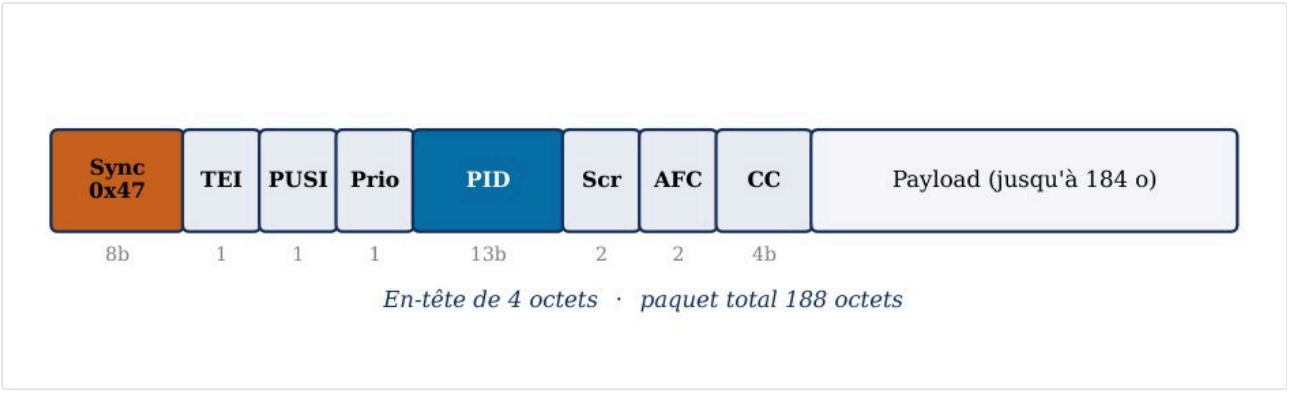


Figure 6.5 — The detailed structure of an MPEG-2 TS packet (sync 0x47, PID, scrambling, continuity counter…).

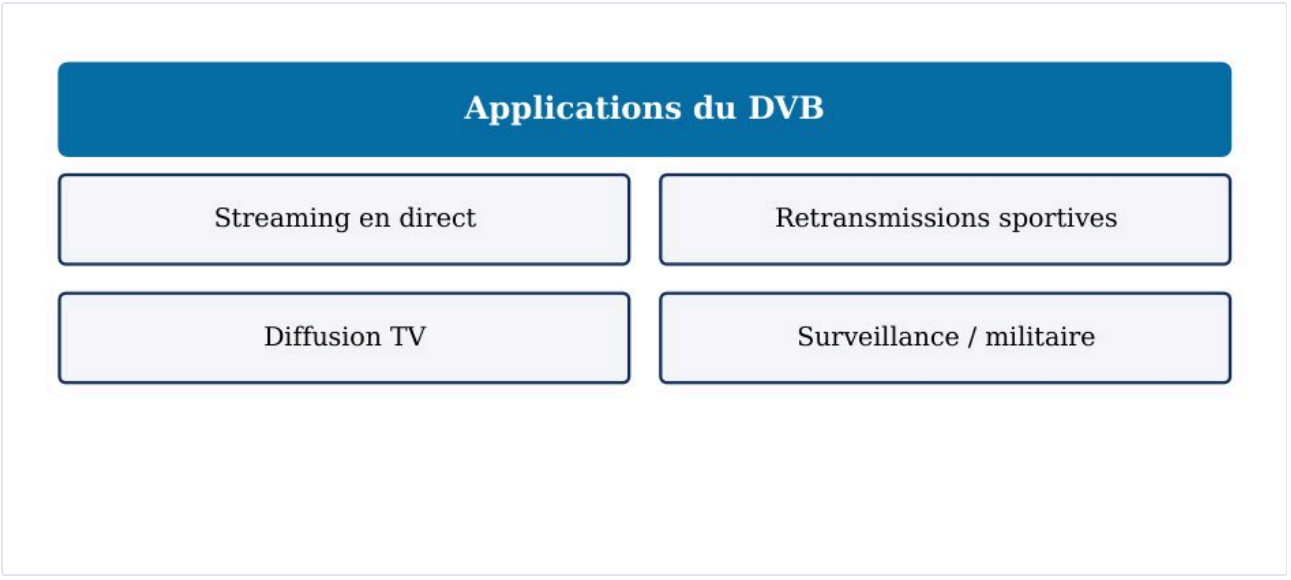


Figure 6.6 — Typical DVB uses (live streaming, sports feeds, surveillance).

6.3 Satellite modems and Internet access

A SAT modem encodes/decodes data for transmission; its intermediate-frequency output is modulated and, on the ground, down-converted by a LNB (Low-Noise Block downconverter).

One-way IP link. The single-channel SAT modem uses the DVB-IP protocol: the uplink is a low-rate connection (often tunnelled over the Internet) to the provider, while the downlink delivers the service. An attacker can sniff the downlink addressed to the client.

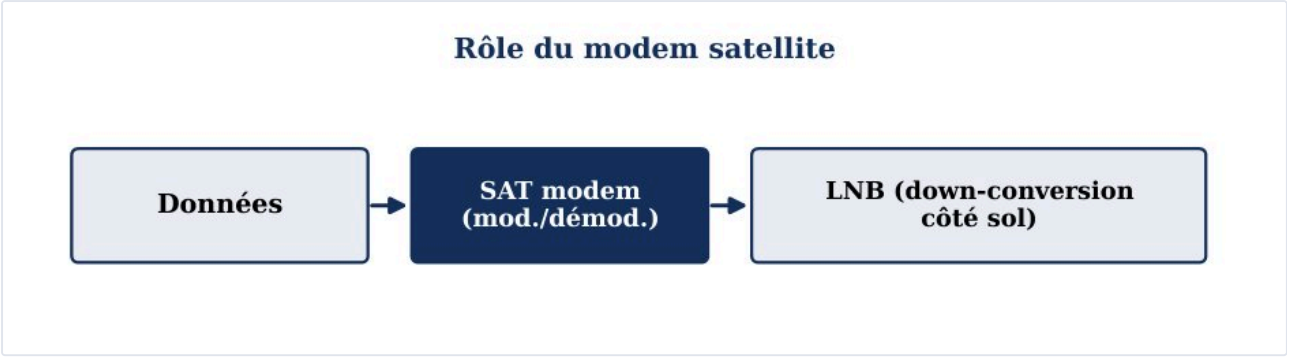


Figure 6.7 — The role of the satellite modem and LNB down-conversion.

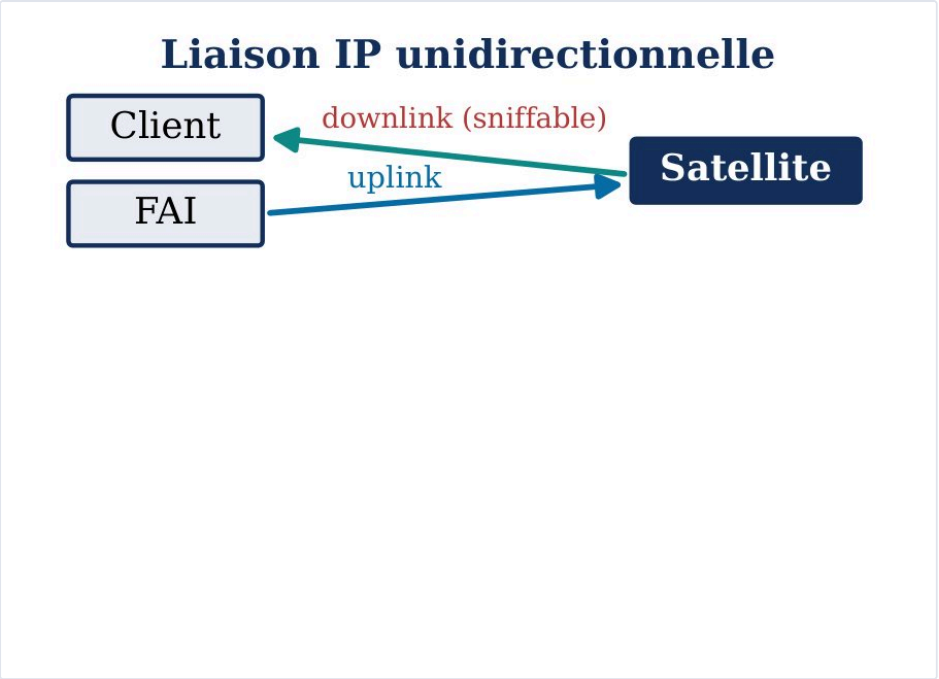


Figure 6.8 — One-way IP link and the eavesdropping point on the downlink.

Two-way link (Astromodem). The Astromodem provides two-way Internet access via the DVB-RCS protocol (Return Channel via Satellite). The client talks directly to the satellite, with no Internet tunnel: both uplink and downlink can be listened to.

6.4 Sniffing DVB: the toolchain

Since DVB is an open and often unencrypted protocol, its packets can be captured. The minimal tooling: a tuned antenna, a LNB, a DVB PCI card (DVB-S/S2), a software DVB analyser, GNU Radio and Wireshark.

1. **Tune the DVB card** — set the transponder parameters in a channels.conf file (values obtained from a service like satfinder), then run szap/tzap.
2. **Find the PID** — using dvbscan and dvbsnoop.
3. **Bring up an Ethernet interface** — dvbnet -a <adapter> <PID>, then ifconfig dvb0_<n> up.
4. **Capture** — with wireshark or tshark on the DVB interface created.

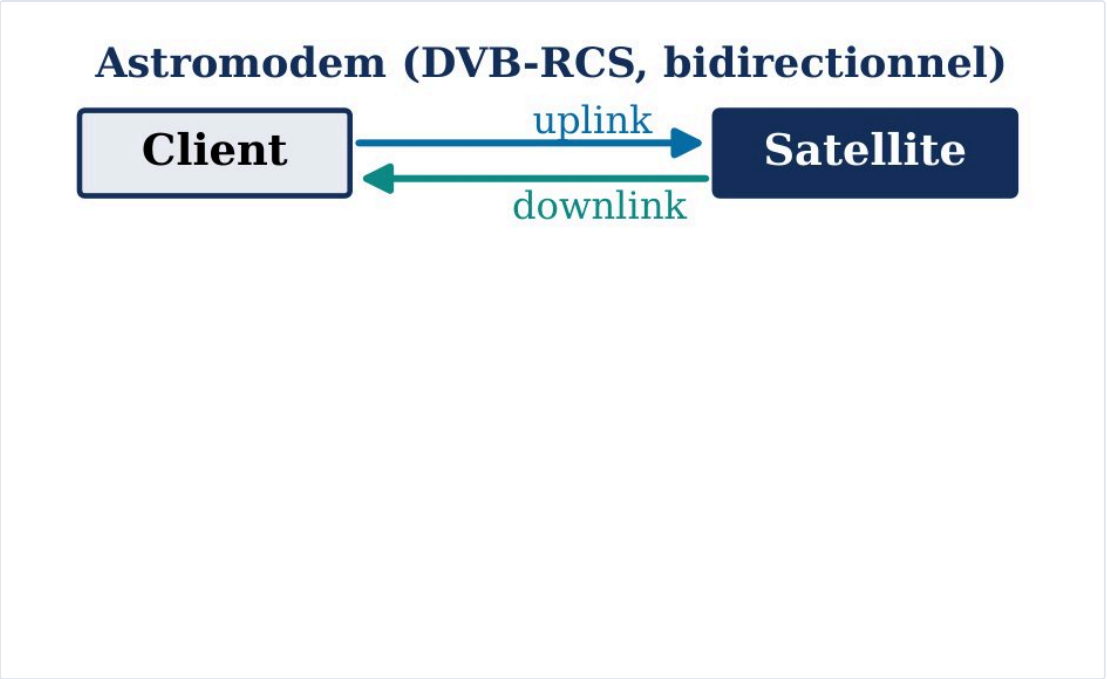


Figure 6.9 — The two-way DVB-RCS link (Astromodem) and eavesdropping surfaces.

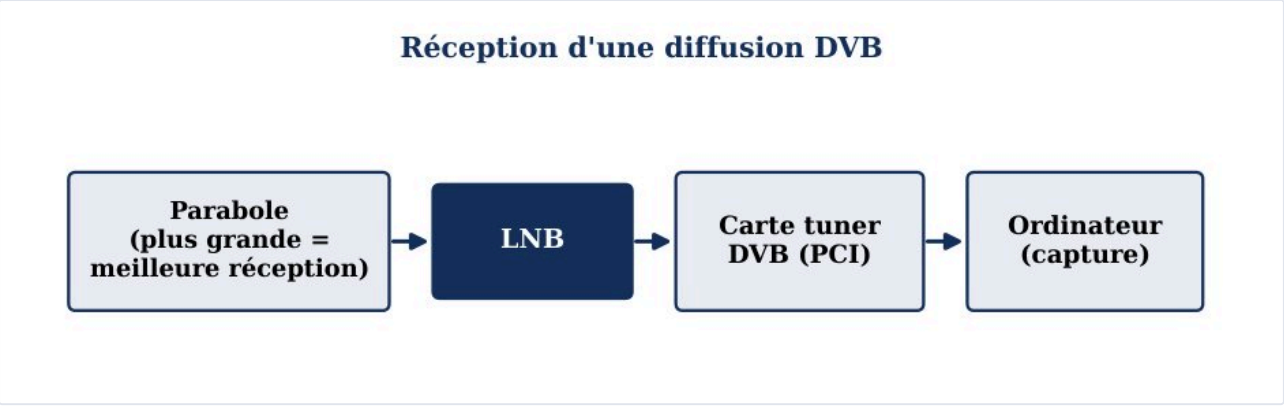


Figure 6.10 — Equipment for listening to a DVB broadcast (dish, LNB, DVB tuner card).

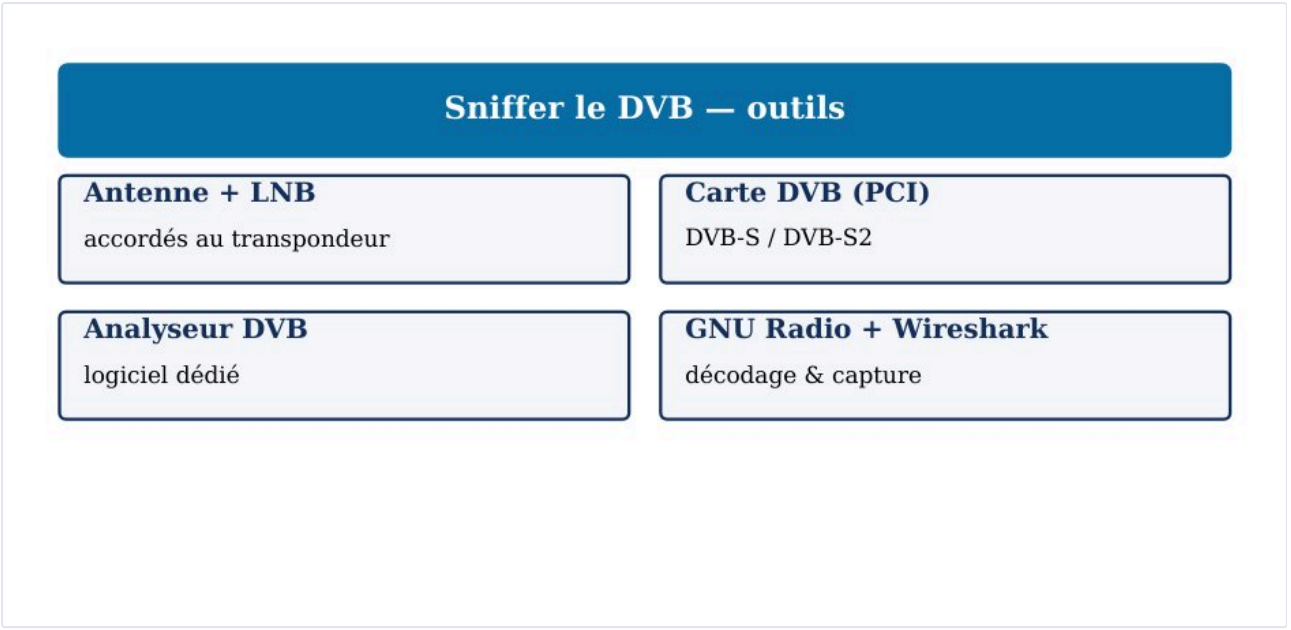


Figure 6.11 — Hardware and software needed to capture DVB packets.

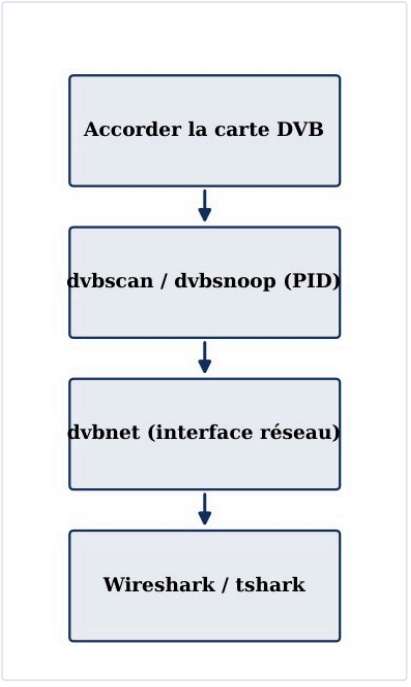


Figure 6.12 — An overview of the DVB listening procedure.

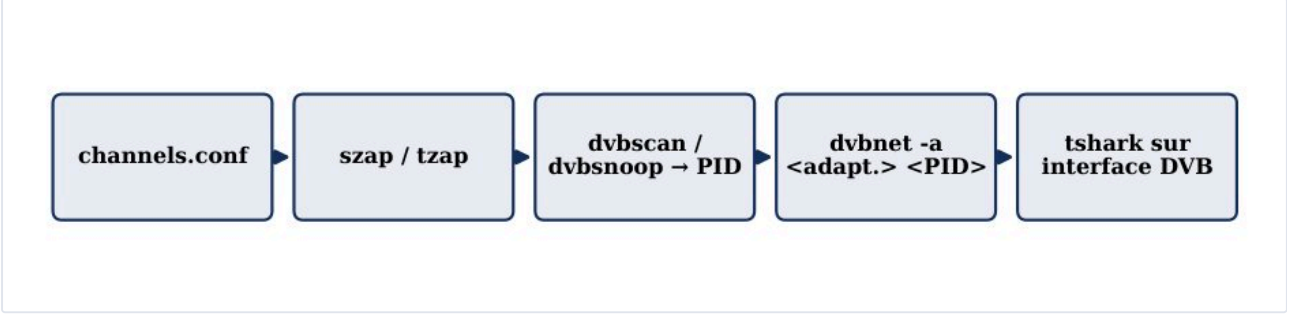


Figure 6.13 — The chain of tools (tune, PID, interface, capture).

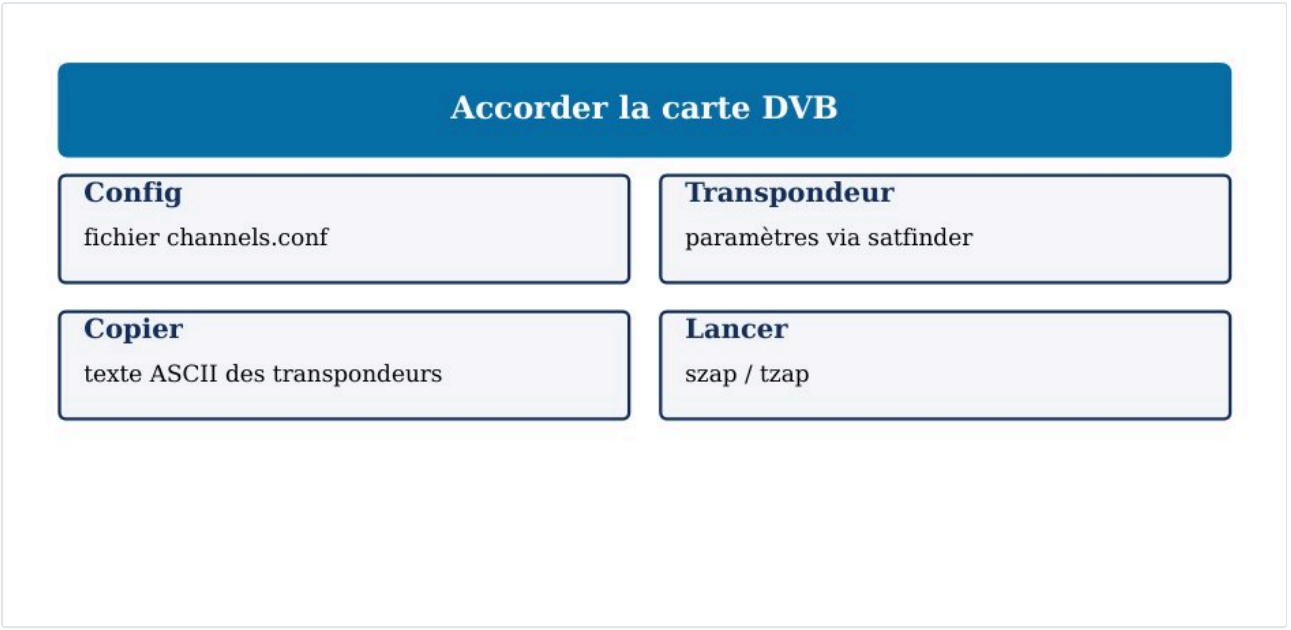


Figure 6.14 — Setting the transponder parameters and the configuration file.

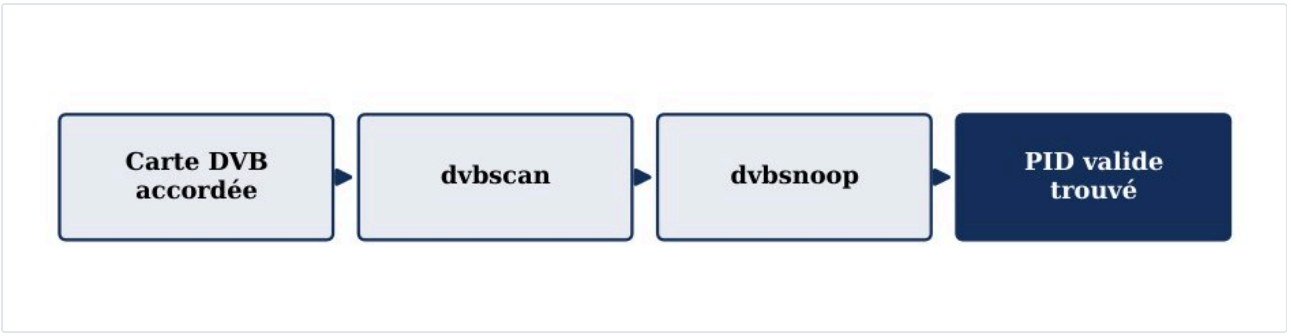


Figure 6.15 — Finding the PID with dvbsnoop/dvbscan.

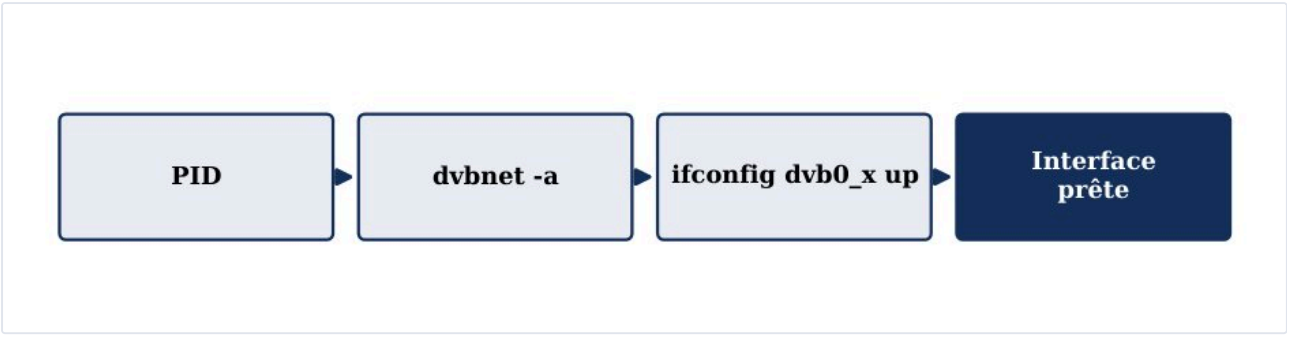


Figure 6.16 — Creating the DVB network interface via dvbnet.

6.5 GNU Radio and the gr-dvbs2rx module

GNU Radio is a generic RF-analysis and signals-intelligence environment in which you assemble processing chains. The gr-dvbs2rx module [10] is a DVB-S2 receiver/transmitter extension: it takes IQ samples as input and outputs an MPEG transport stream (physical-layer synchronisation, baseband-frame processing). Docker is used to isolate the environment.

Key notions. TX/RX are transmit and receive; the transport stream is the multimedia container; the IQ stream is the complex signal (I + jQ). In typical use, you download a sample transport stream (wget/curl), then the dvbs2-tx transmitter reads that TS and produces an IQ stream to the dvbs2-rx receiver, which can show a GUI (--gui option).

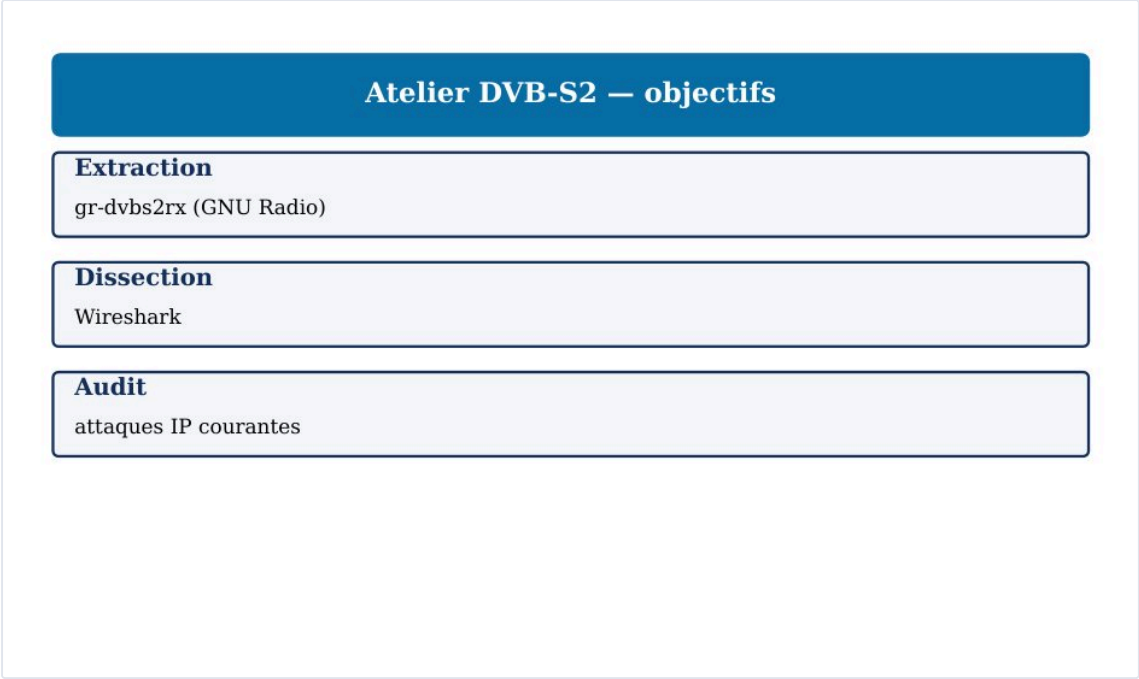


Figure 6.18 — GNU Radio as a generic RF-analysis tool.

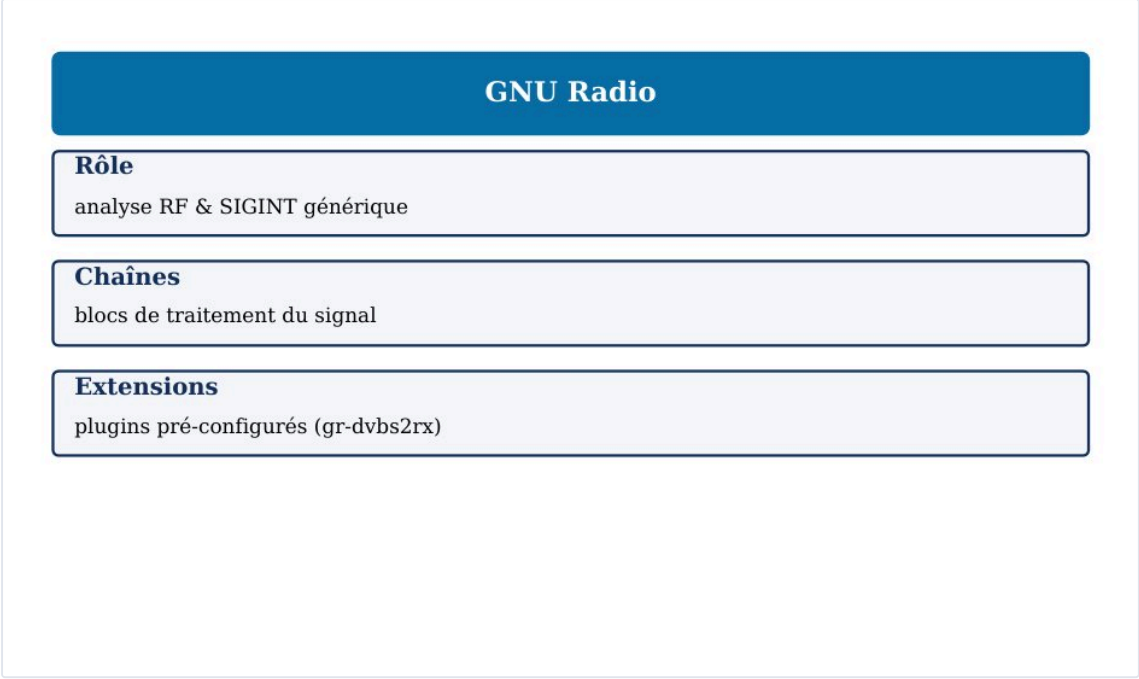


Figure 6.19 — GNU Radio: role, processing chains and extensions (gr-dvbs2rx).

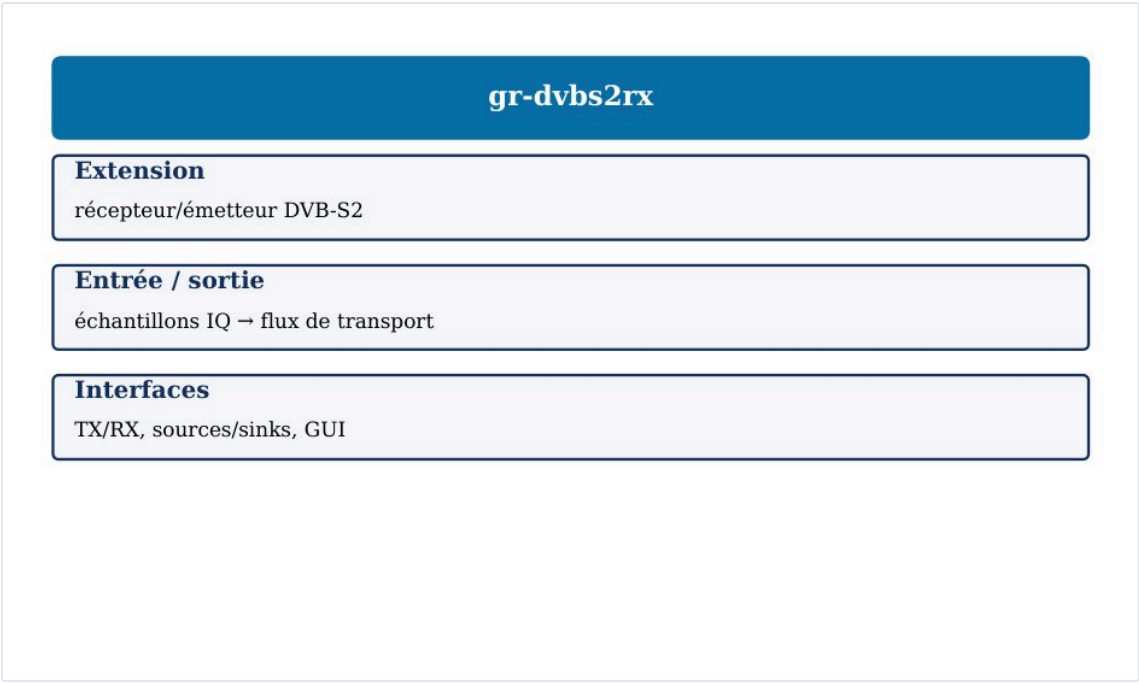


Figure 6.20 — The DVB-S2 receiver extension for GNU Radio (TX/RX, sources/sinks, interface).

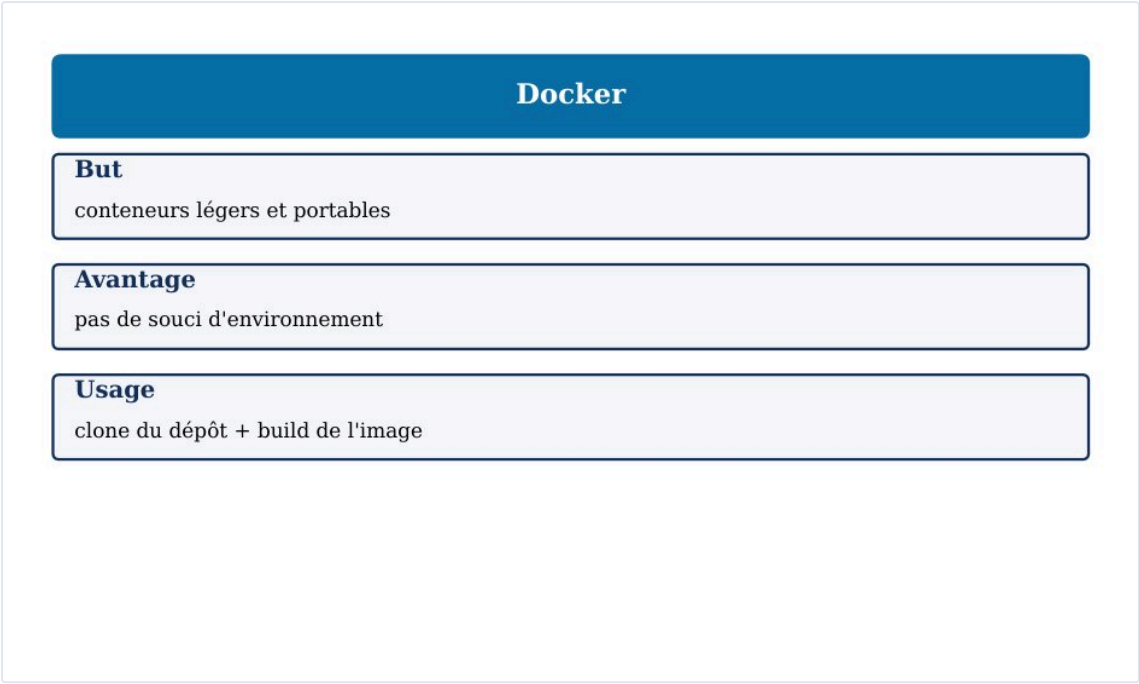


Figure 6.21 — Using Docker to deploy the gr-dvbs2rx environment.



Figure 6.22 — Cloning the repository and building the gr-dvbs2rx image.

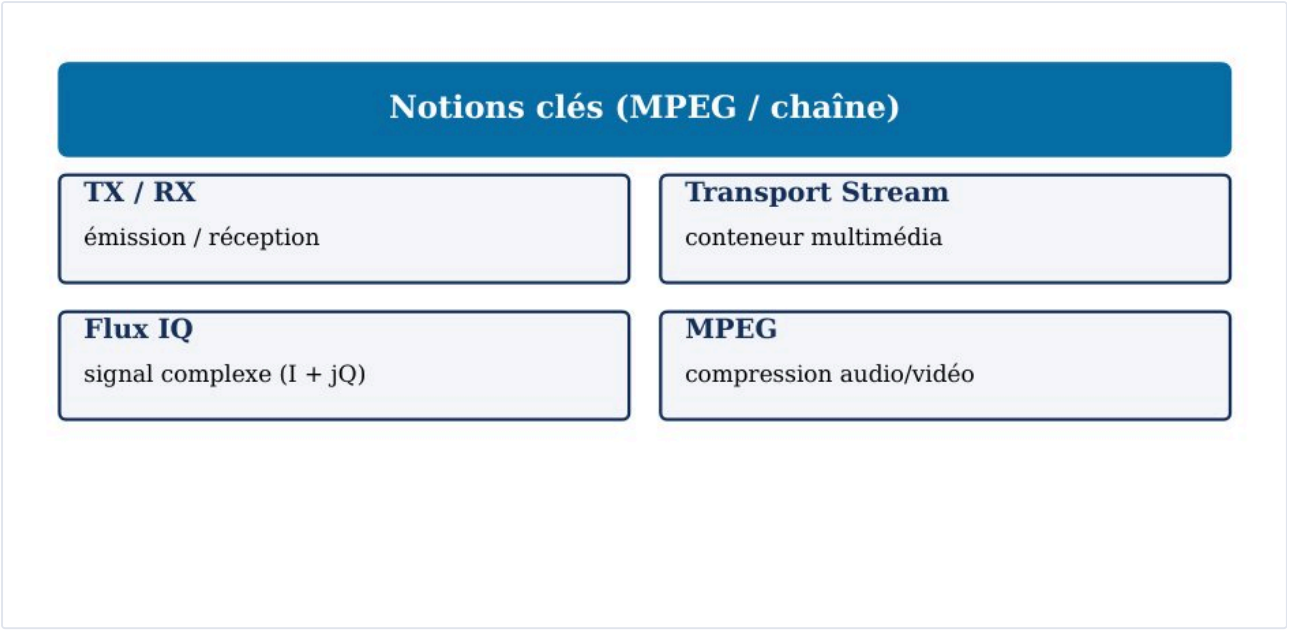


Figure 6.23 — A reminder of the MPEG format and the TX/RX, transport-stream and IQ notions.

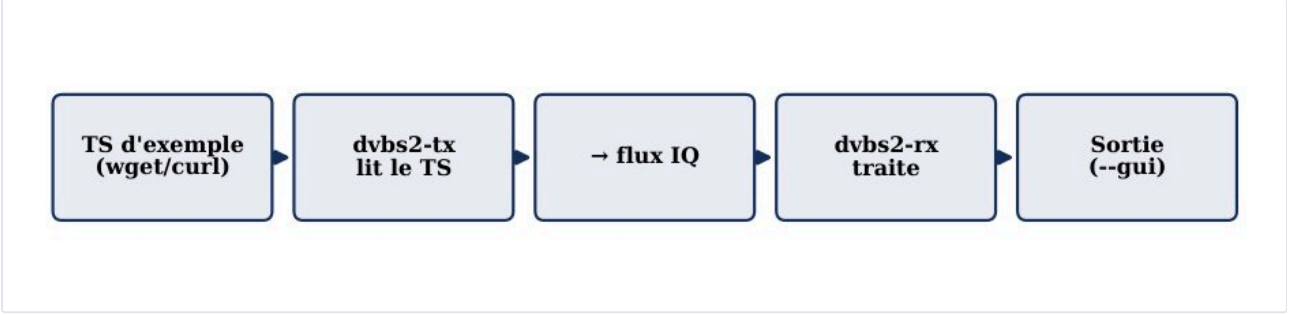


Figure 6.24 — Usage commands (downloading a sample TS, the tx/rx chain).

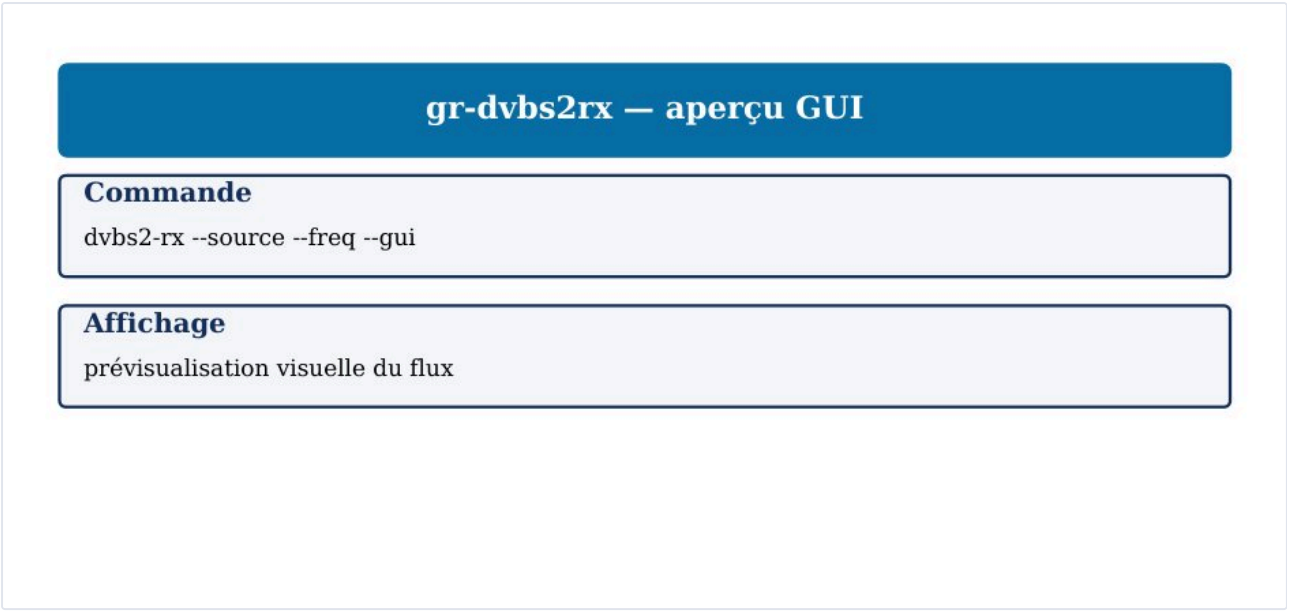


Figure 6.25 — A preview of the DVB-S2 receiver's GUI.

6.6 Dissecting communications with Wireshark

Wireshark [12] is an open-source network-protocol analyser. For DVB-S2, you load a PCAP file containing DVB traces, then enable the dissector: Analyze → Enabled Protocols, search “DVB”, tick “DVB-S2 UDP”. You then filter the baseband frames (BBFrame) of the DVB-S2 stream over UDP.

DVB-S2 baseband frame. The baseband frame (BBFrame) is the output of the DVB-S2 receiving equipment. The Wireshark dissector relies on UDP for transport; the baseband payload is encapsulated in GSE (Generic Stream Encapsulation) and/or MPE (IP over DVB). Wireshark uses a heuristic dissector: it pre-reads each packet to decide whether it matches the DVB-S2-over-UDP criteria [13]. The adaptation-mode header carries the modulation scheme, the FEC, and so on; the BBFrame section carries user data, control and signalling information.

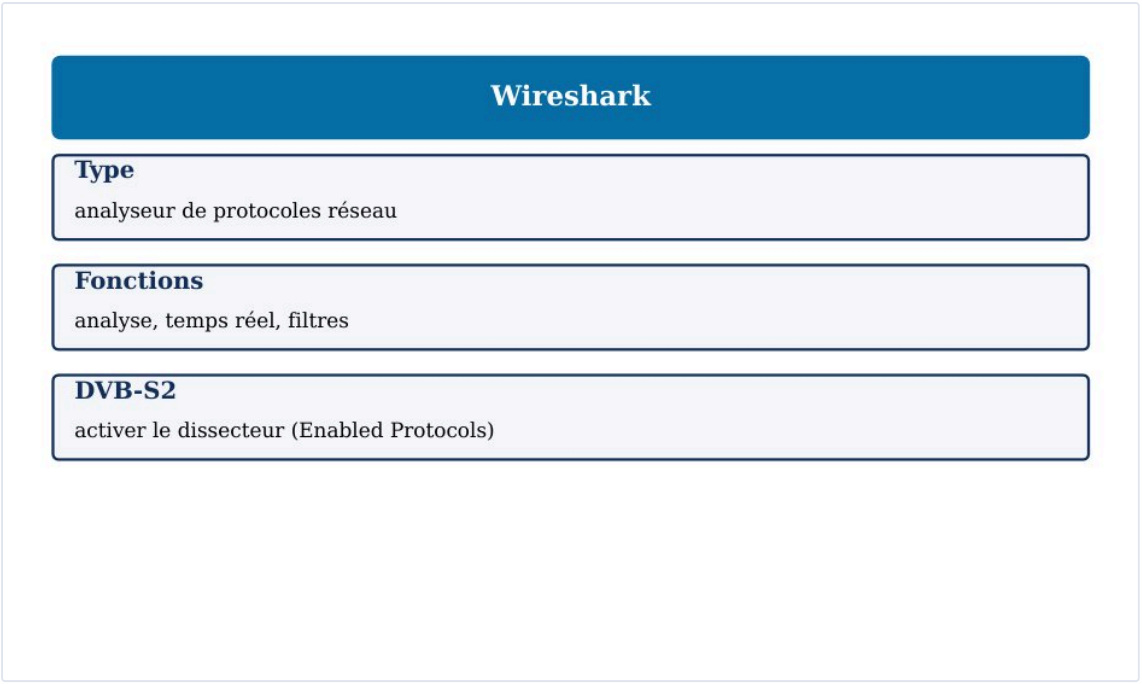


Figure 6.26 — An overview of Wireshark and DVB-S2 packet analysis.

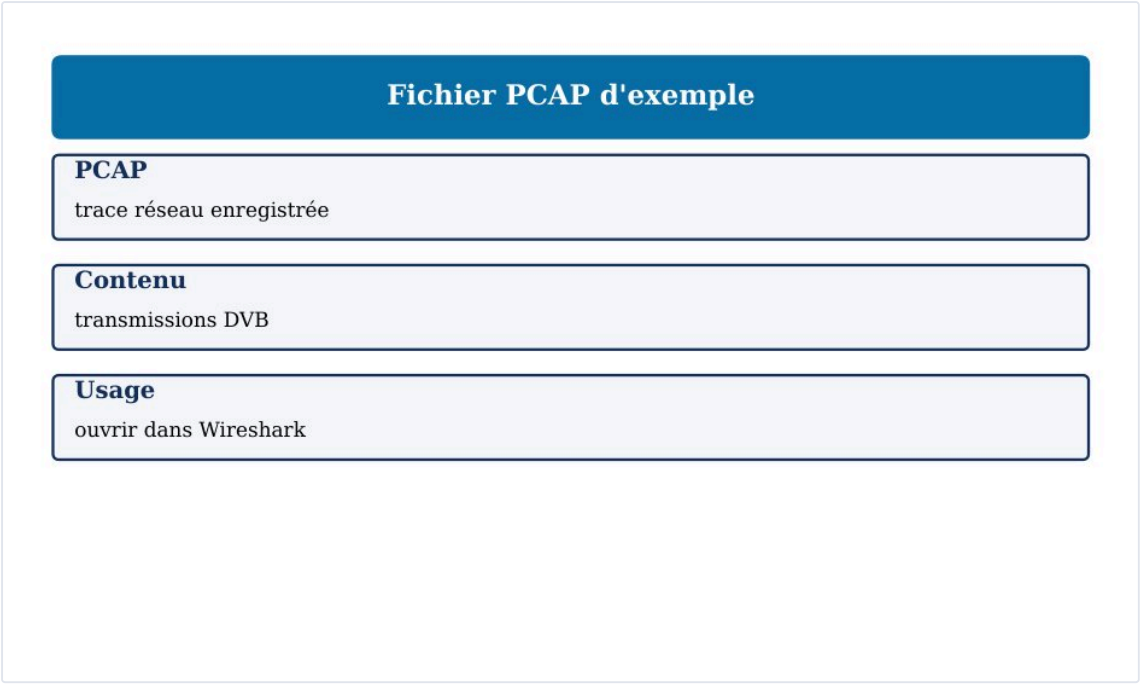


Figure 6.27 — Fetching the sample capture file containing DVB traces.

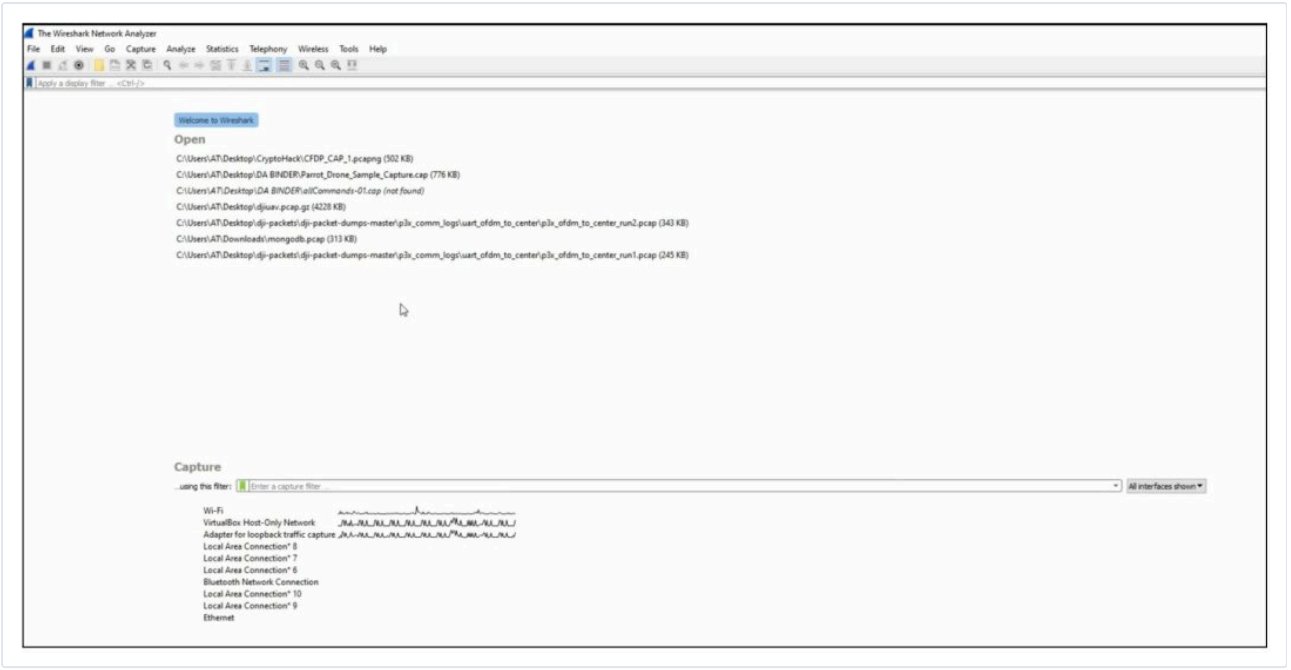


Figure 6.28 — Wireshark: a DVB-S2 UDP stream opened in the analyser.

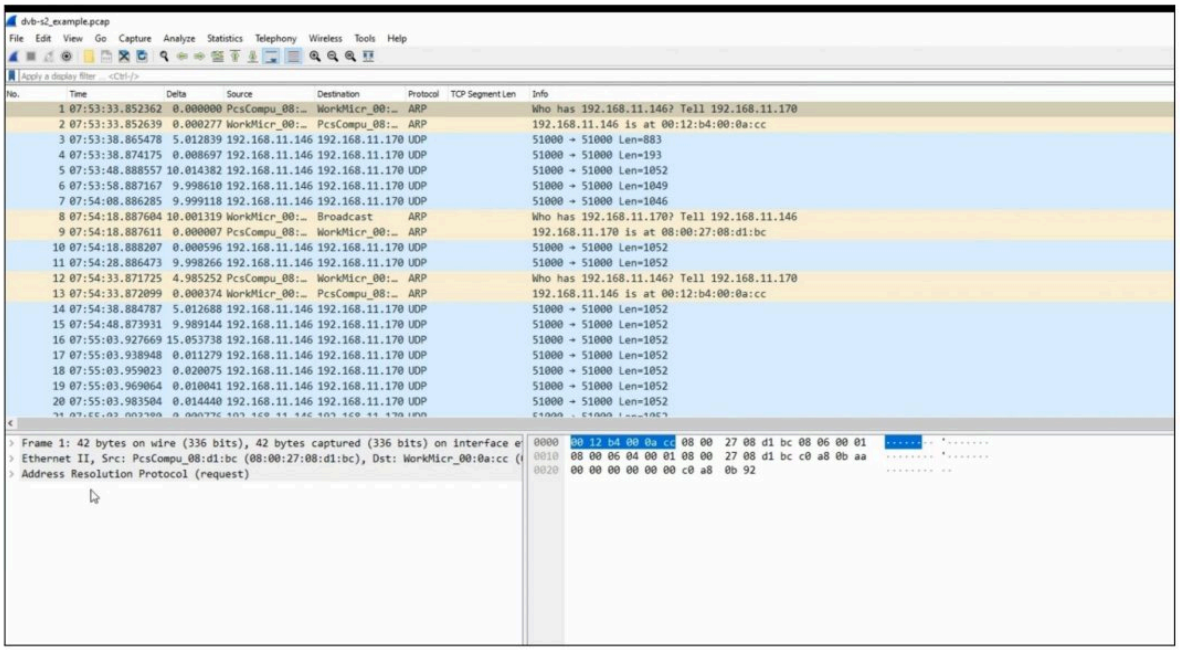


Figure 6.29 — Wireshark: enabling/filtering the DVB-S2 dissector.

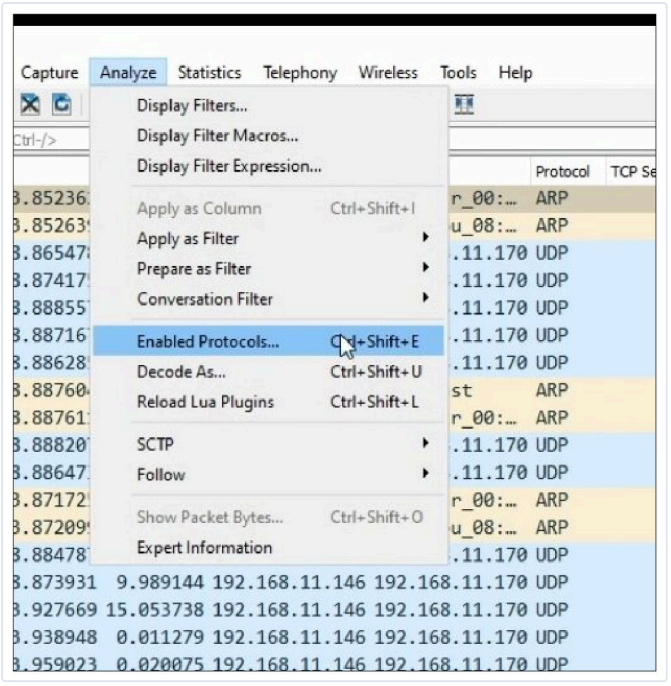


Figure 6.30 — Wireshark: the list of DVB-S2 baseband frames.

6.7 Network threats specific to DVB

The main threats against DVB satellites take the form of specialised network attacks:

- **DNS spoofing** — a misconfigured DNS entry points to another IP, “poisoning” the name server and misdirecting the receiver's session;
- **TCP hijacking** — an attacker takes over a TCP session;
- **GRE packet attacks** — exploiting the GRE protocol, intrinsic to some DVB satellites, for unauthorised access.

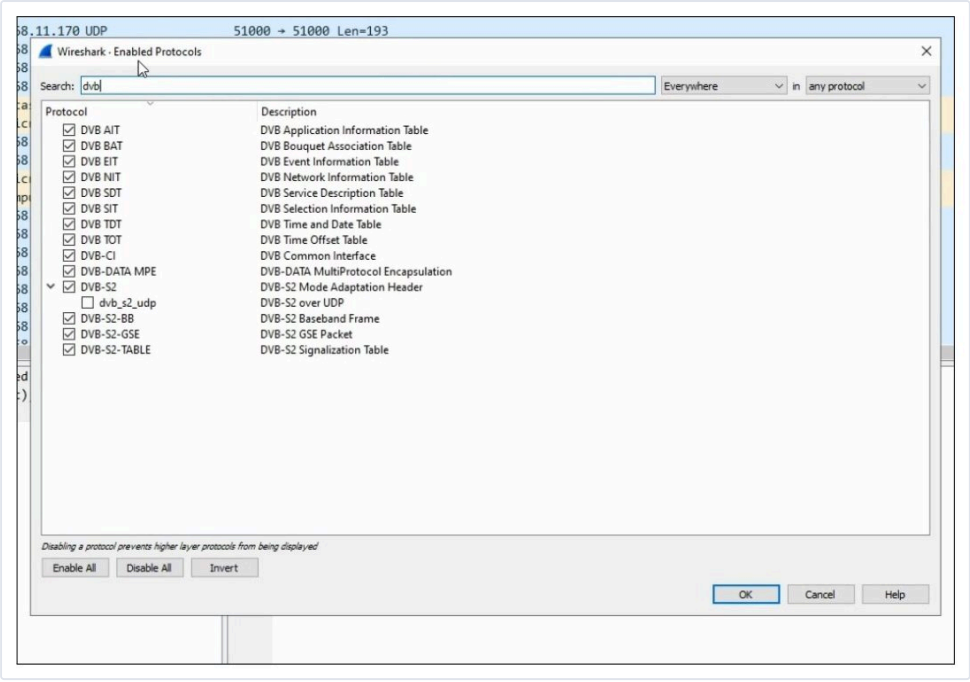


Figure 6.31 — Wireshark: frame detail (length, DFL, sync, checksum, data).

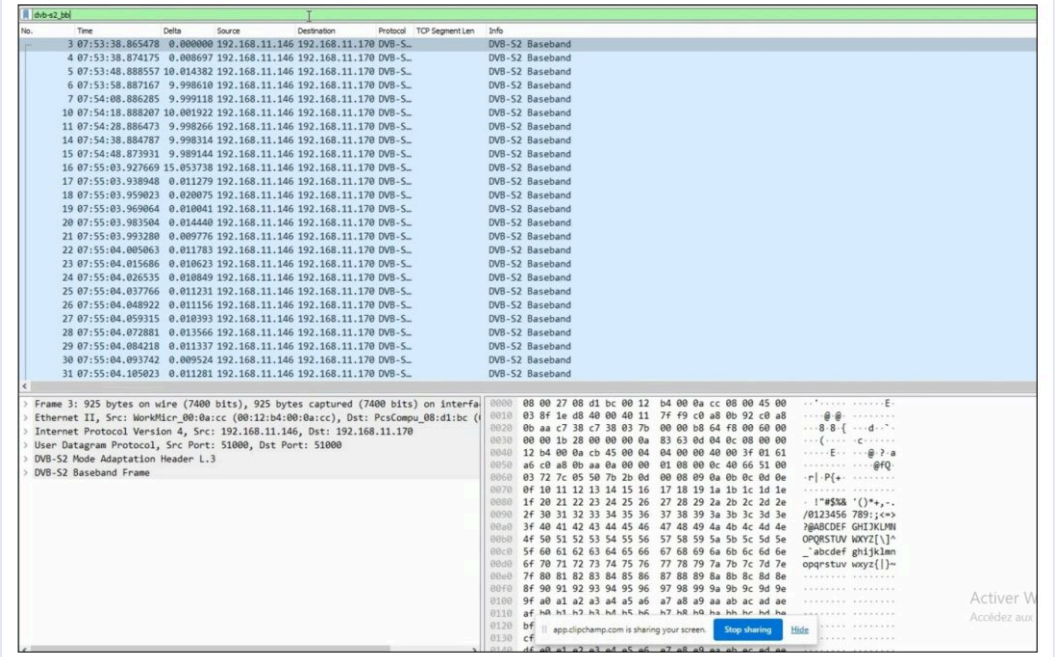


Figure 6.32 — Wireshark: inspecting the baseband-frame fields.

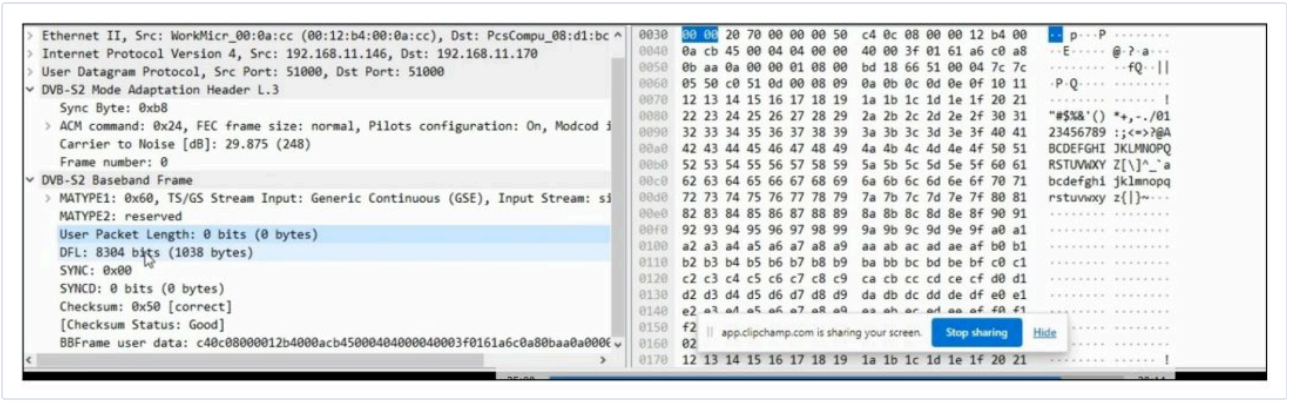


Figure 6.33 — Wireshark: continuing the dissection of DVB-S2 packets.

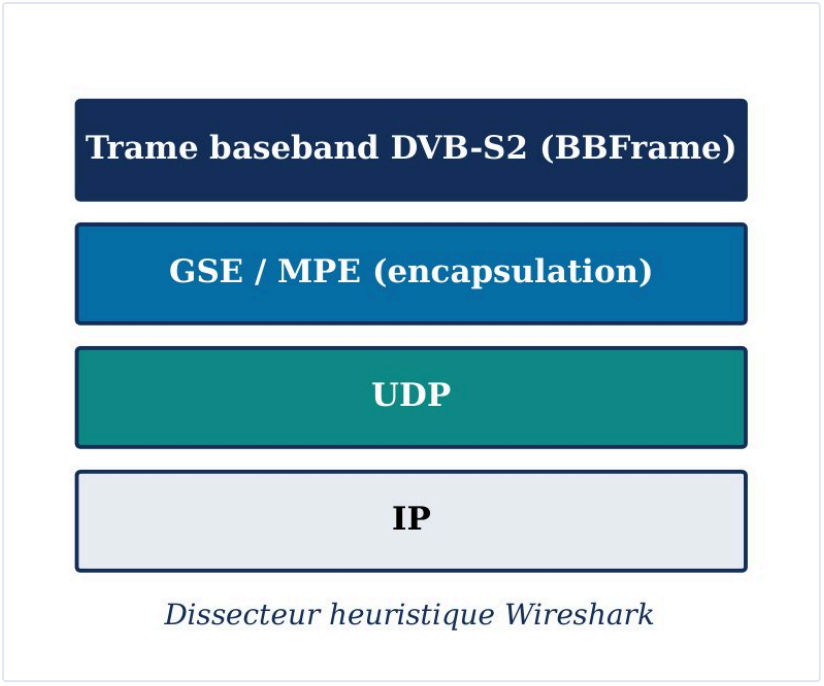


Figure 6.34 — The baseband-frame composition (UDP dependency, GSE/MPE payload, heuristic dissector).

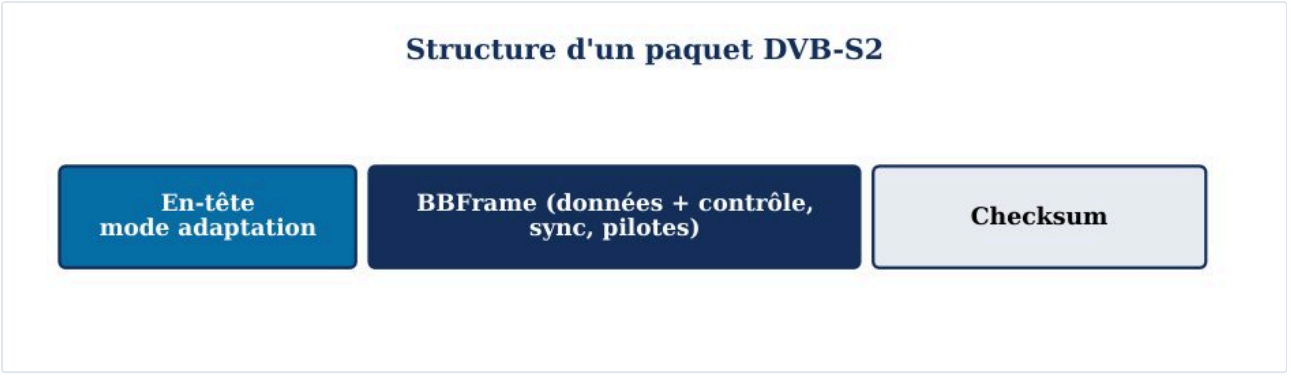


Figure 6.35 — The full structure of a DVB-S2 packet (header, BBFrame, sync, pilots, checksum).

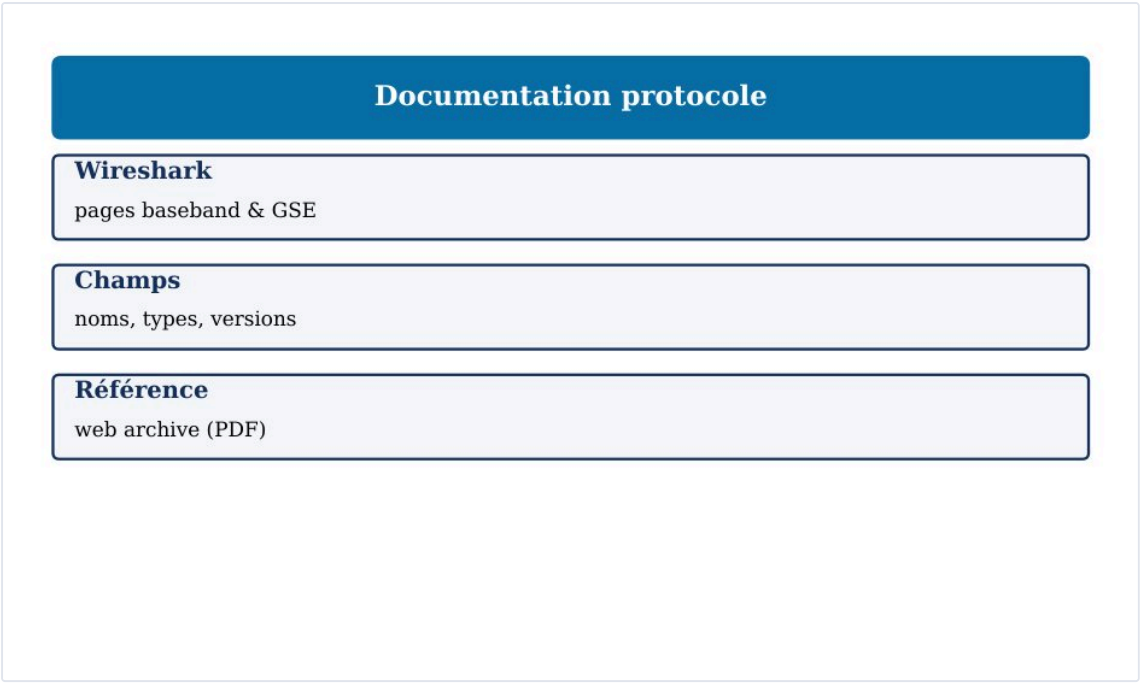


Figure 6.36 — Pointers to the Wireshark documentation for the baseband and GSE protocols.

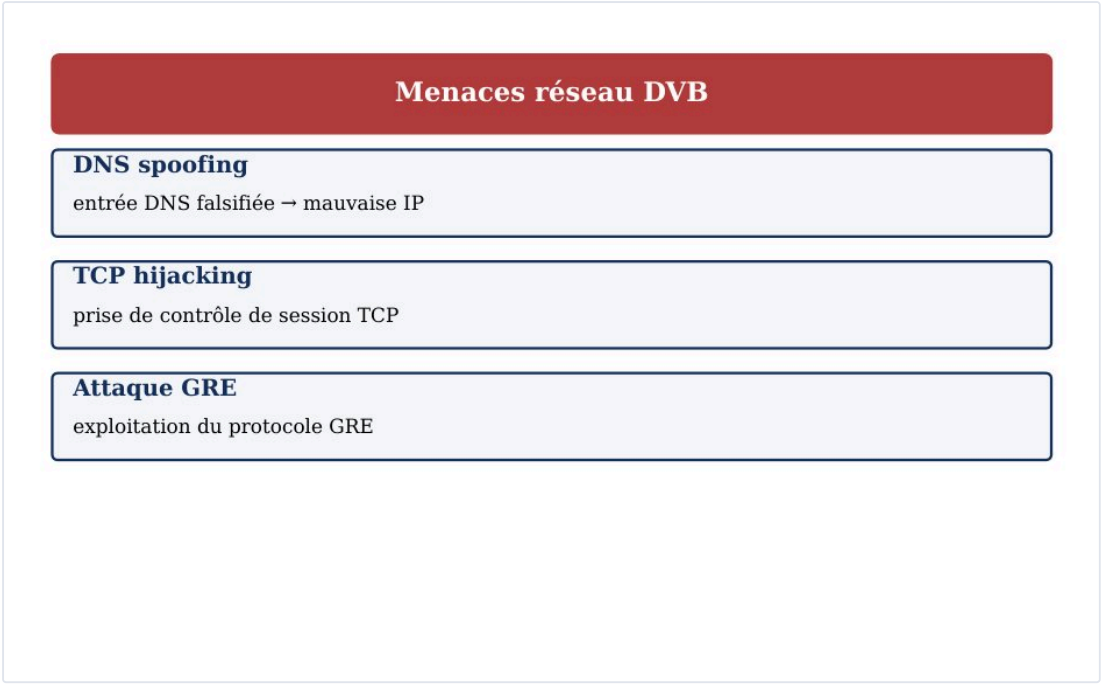


Figure 6.37 — Three major network threats: DNS spoofing, TCP hijacking, GRE attack.

CHAPTER 7

SatNOGS and telemetry collection

7.1 The SatNOGS network

SatNOGS (Satellite Networked Open Ground Station) [5] is a global open-source network of ground stations, a Libre Space Foundation project. It has four sub-projects: Network (a web app for scheduling observations), Database (a community transmitter database with an API), Client (the ground-station software) and Ground Station (open hardware). The network has more than 200 operational stations and observations from more than 400 satellites.

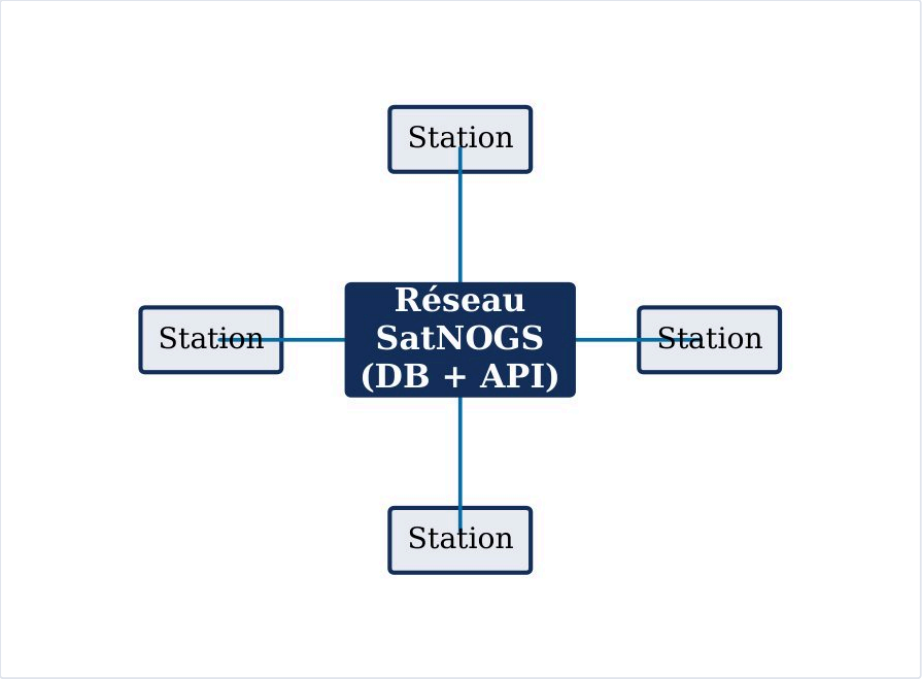


Figure 7.1 — A global network of ground stations, a database and an API.

Dashboards. A central feature is the suite of dashboards, each gathering a satellite's telemetry time series: decoded messages, frames and beacons, status messages and subsystem data.

7.2 The vocabulary of SatNOGS pages

SatNOGS listings, mostly amateur-radio (AMSAT), use a specific vocabulary: **Digipeater** (an amateur digital repeater that receives, processes and re-transmits data); **VHF/UHF TRX** (transceiver); **APRS** (short digital messages over the Automatic Packet Reporting System, widely used in emergency services); **PSU** (power-supply telemetry), **OBC** (on-board computer) and **BDS** (navigation data).

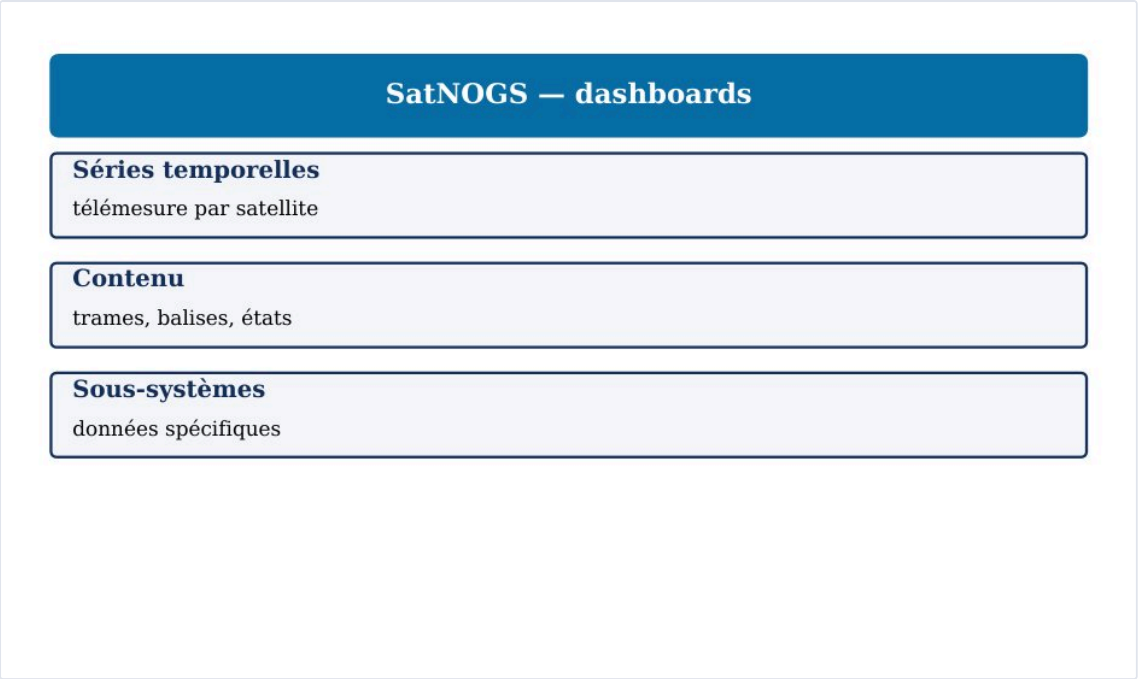


Figure 7.2 — A satellite's telemetry time series and subsystem data.

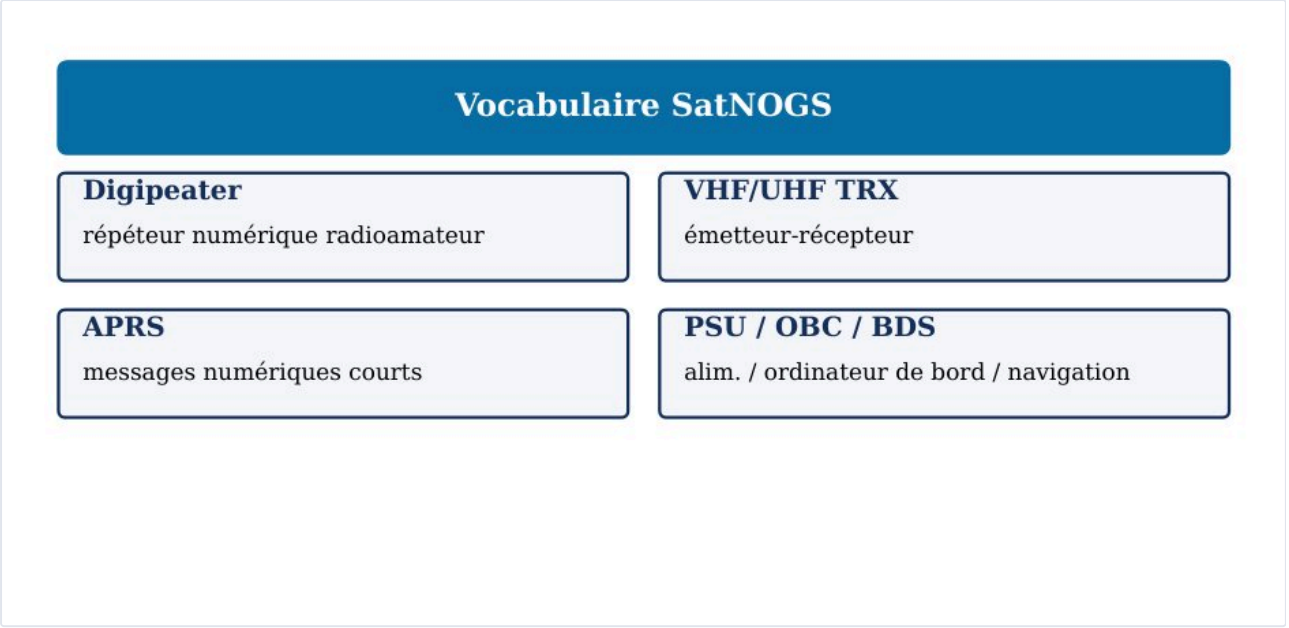


Figure 7.3 — SatNOGS vocabulary: digipeater, TRX, APRS, PSU/OBC telemetry…

7.3 Case study: the BDSAT-2 satellite

As an example, we retrieve the telemetry of BDSAT-2, a small CubeSat listed by SatNOGS, via its dedicated dashboard. We observe: the digipeater messages (VHF and UHF) with the call signs of the latest operators, the count and timestamps of the telemetry frames, the receiving station and a ranking of stations. APRS messages (ASCII) are decoded with an APRS decoder; PSU telemetry gives battery levels, currents, temperatures and channel states; OBC telemetry gives solar temperatures and uptime. This data is valuable for the reconnaissance (OSINT) phase on a small satellite.

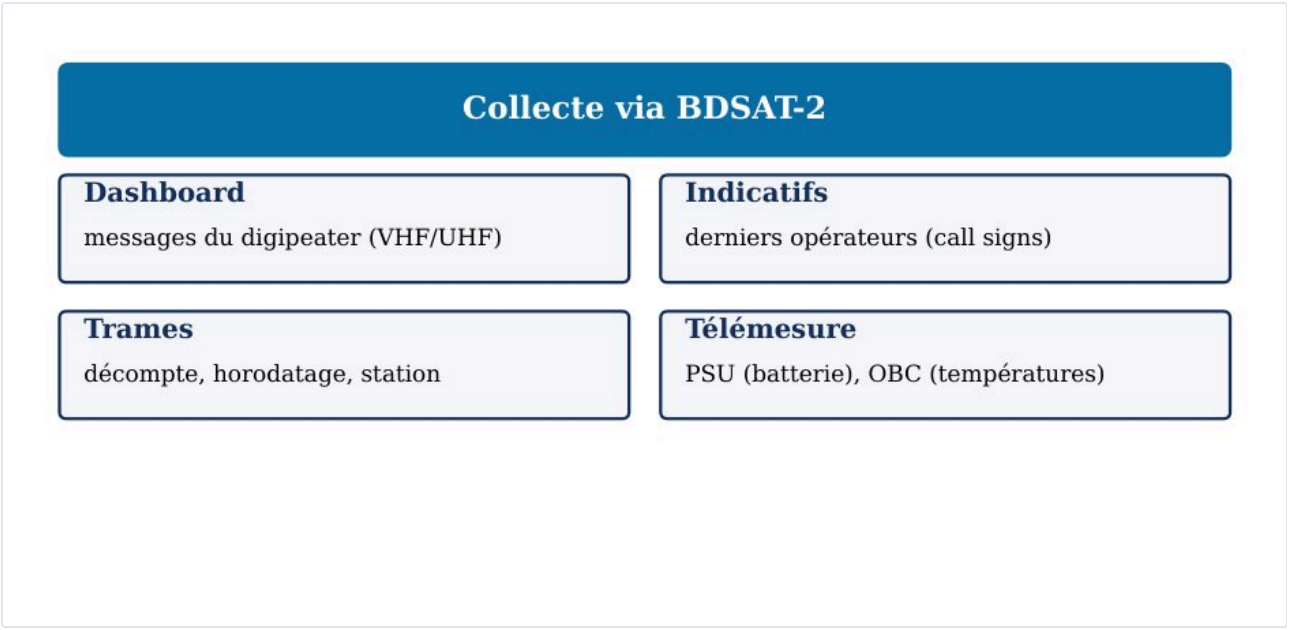


Figure 7.4 — Using the SatNOGS dashboard to collect BDSAT-2 CubeSat telemetry.

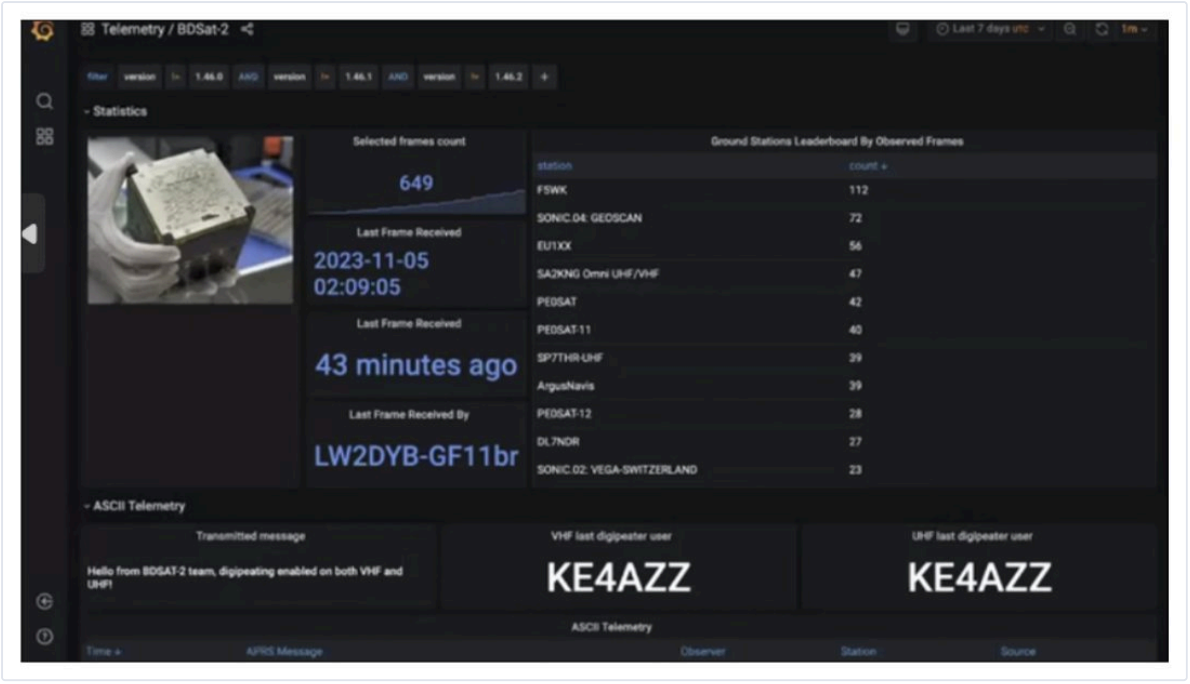


Figure 7.5 — A dashboard view: digipeater messages, call signs, frames, stations, subsystem telemetry.

CHAPTER 8

Reverse engineering of satellite communications

8.1 Levels of reverse engineering

Reverse-engineering a satellite communication is approached in levels, mirroring the OSI model:

- **Physical layer** — the main goal: convert waveforms into digital data. Techniques: digital-modulation recognition (via tools exploiting amplitude, frequency and phase) and identification of the transmission technique (MIMO, Direct-Sequence Spread Spectrum).
- **Link and network layers** — frame synchronisation via predefined patterns, channel-code recognition (linear block vs convolutional codes), reconstruction of interleavers and scramblers.
- **Application/session/presentation layers** — access to multimedia data (video, photos, audio) by identifying the source coders.

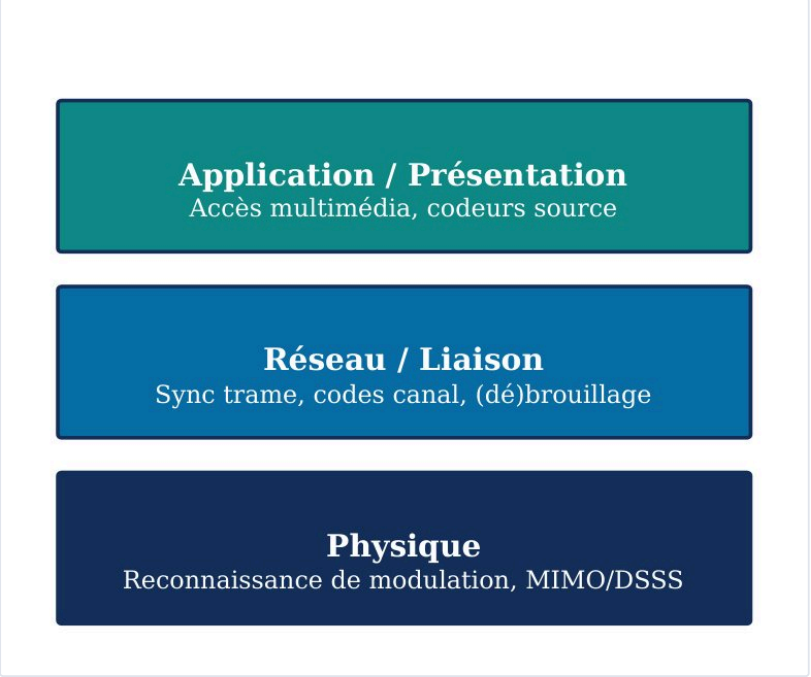


Figure 8.1 — Dimensions of reverse engineering: physical, link/network, application.

8.2 Communication hardware and reconnaissance

Reverse engineering begins by understanding the hardware: transponder, digipeater, RX/TX components, frequency converters and the telemetry, tracking and command (TT&C) subsystem that regulates the satellite/ground dialogue. OSINT reconnaissance aims to gather: band and frequency (uplink/downlink), modulation scheme and error-correcting codes, bandwidth, orbital parameters (via gpredict or N2YO) and antenna systems.

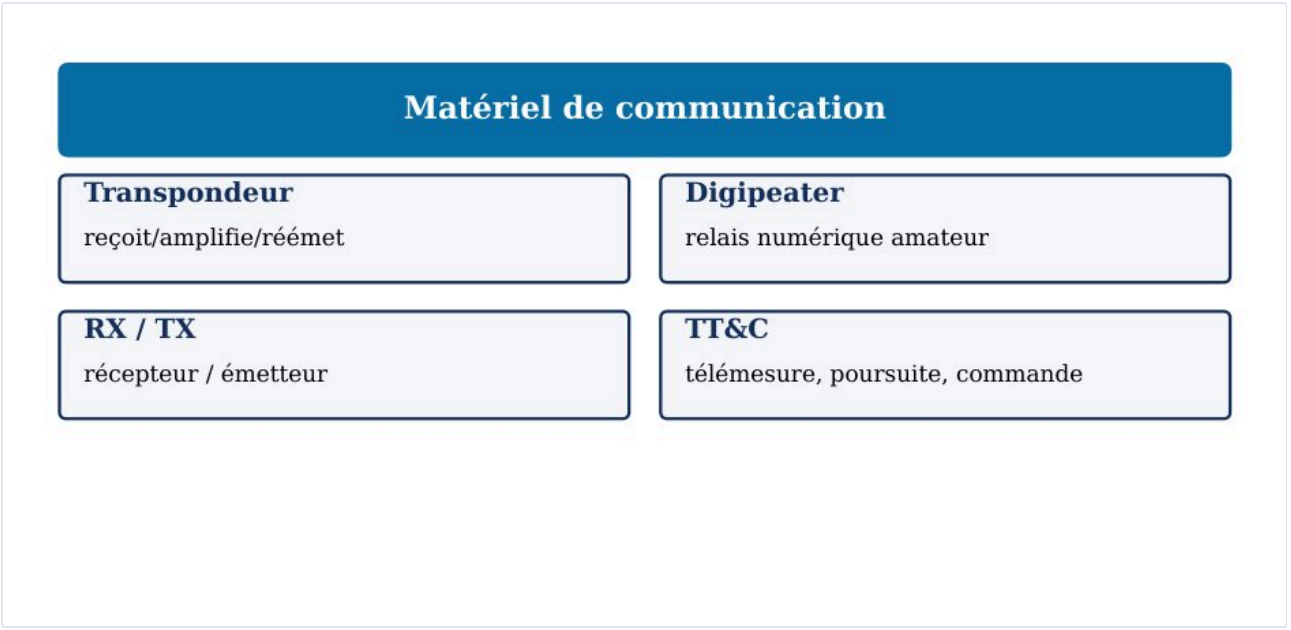


Figure 8.2 — Satellite communication equipment (transponder, digipeater, RX/TX, converters, TT&C).

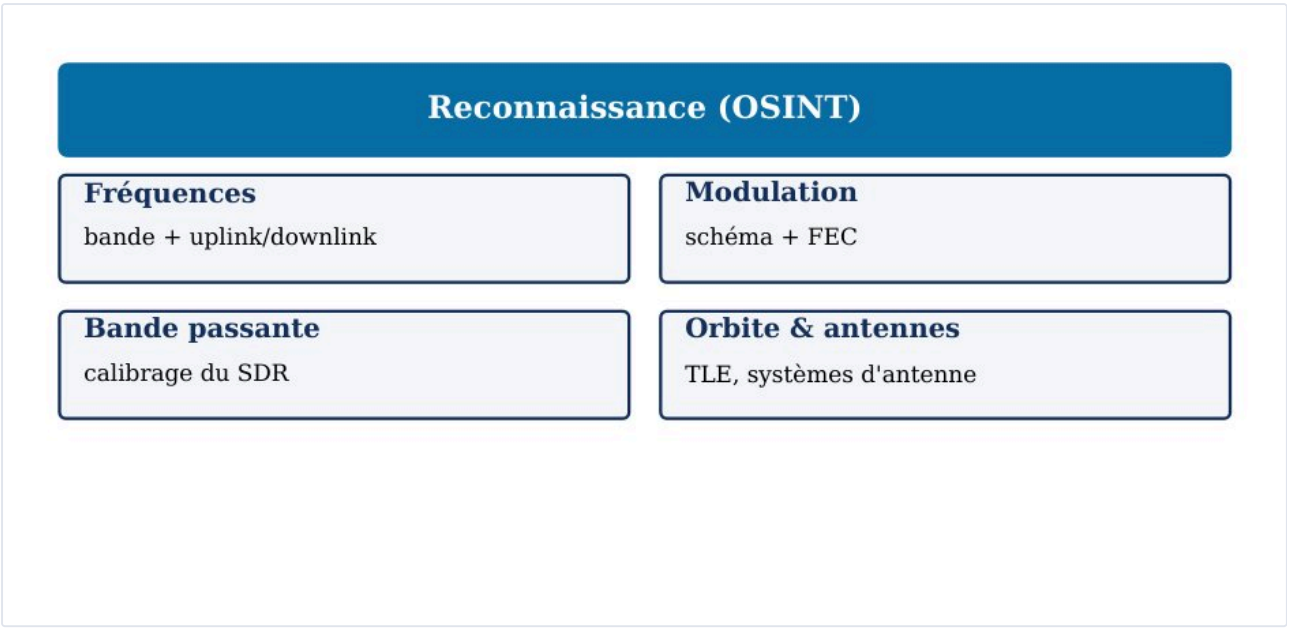


Figure 8.3 — Information to gather by OSINT to calibrate the SDR and acquire the signal.

8.3 Common satellite protocols

Several packet protocols coexist: **CSP** (CubeSat Space Protocol, a network layer designed for CubeSats at Aalborg, adopted by GomSpace [18]); **CCSDS / Space Packet Protocol** (highly standardised and widespread [2]); **APT** (NOAA weather imagery); **DVB-S** (video); **CAN** (Controller Area Network, internal satellite communications); **AX.25** (packets for amateur-radio satellites [19]).

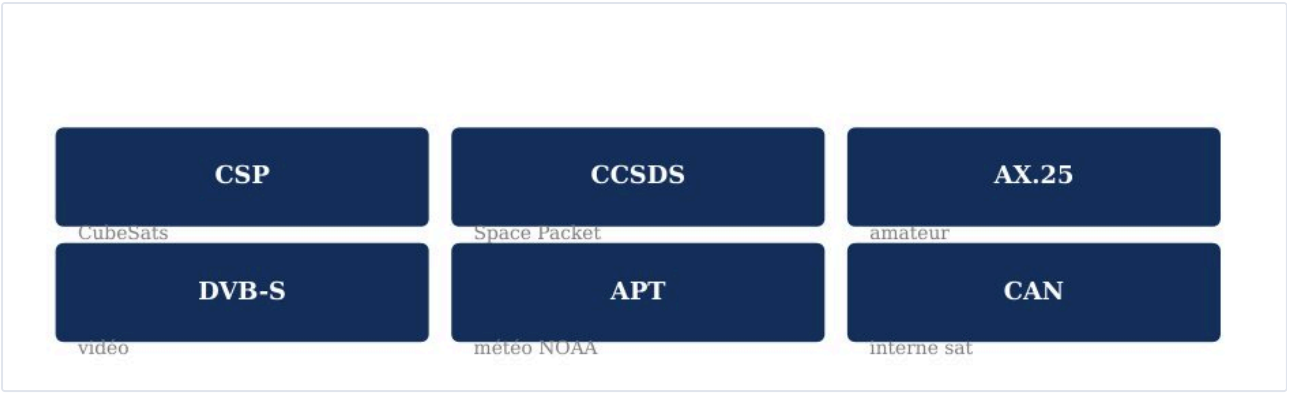


Figure 8.4 — A survey of the protocols (CSP, CCSDS, APT, DVB-S, CAN, AX.25).

8.4 Signal demodulation

Demodulation inspects the modulated carrier to recover the original digital data. You need to know the carrier characteristics (signal-to-noise ratio, synchronisation, modulation index, phase, amplitude — often documented), the modulation technique (AM, FM…) and the error-correction scheme. Two very common codes: Reed-Solomon (block error correction) and the CRC (Cyclic Redundancy Check, error detection).



Figure 8.5 — Demodulation steps and error-correcting codes (Reed-Solomon, CRC).

8.5 From frame decoding to packet analysis

Frame decoding. A frame has three sections: a header (metadata: sync markers, length, control), a payload (useful data, structure depending on the protocol) and error detection/correction codes at the end.

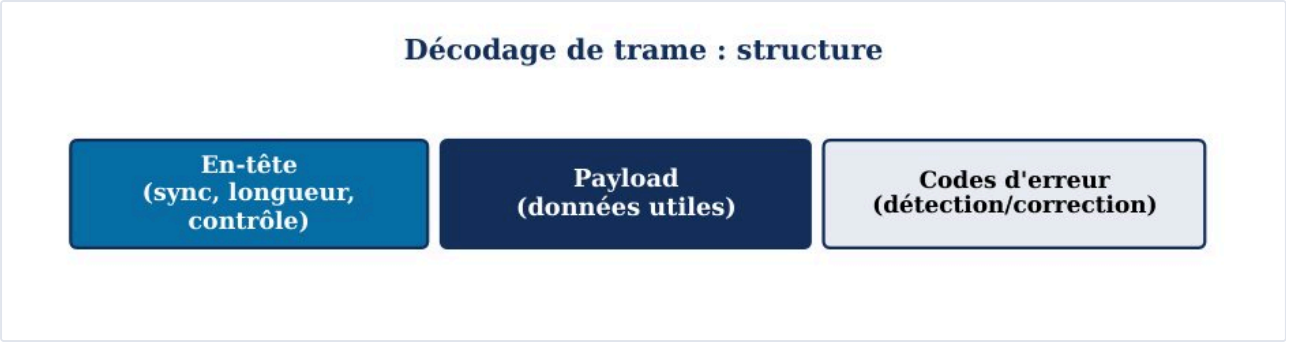


Figure 8.6 — Frame structure: header, payload, error codes.

Packet demuxing. Separating and categorising streams: de-randomising frames (reversing the randomisation, e.g. via a generator polynomial tied to the CRC), demultiplexing virtual channels (by identifiers and headers) and Reed-Solomon / FEC correction for low signal-to-noise ratios.

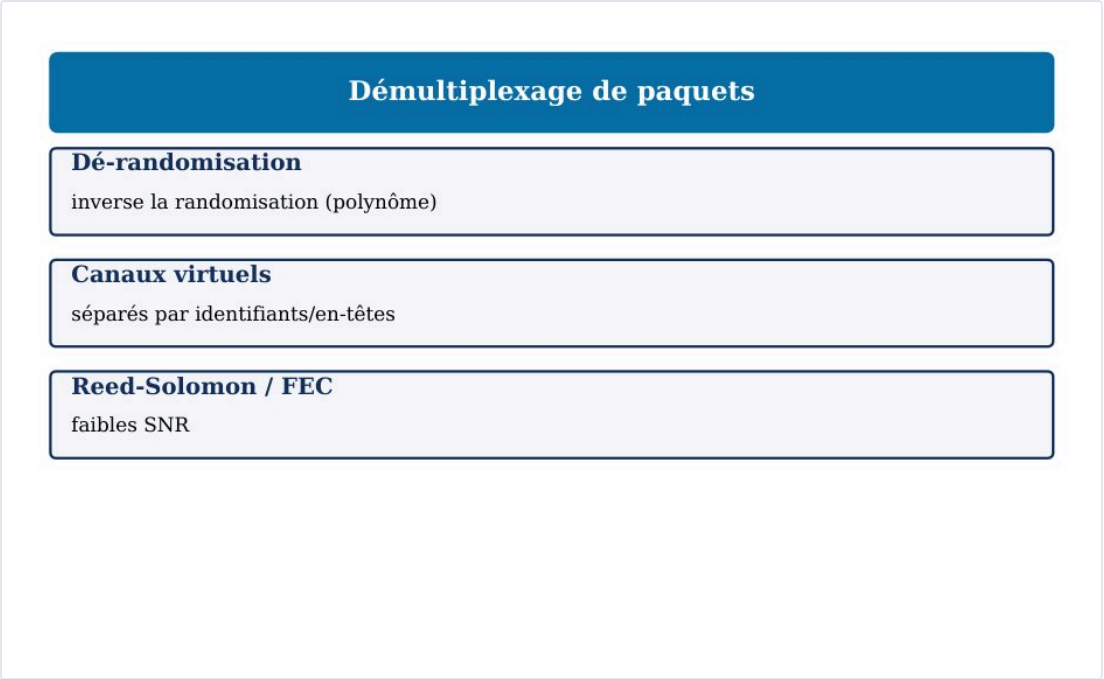


Figure 8.7 — De-randomisation, virtual-channel demultiplexing and FEC.

File assembly. Aggregating and decompressing packets: applying the decompression algorithm and handling continuation flags to reassemble data fragmented across several frames. **Packet analysis.** Understanding the structure and content: knowledge of the protocol (CCSDS defines precise rules), reading header flags, exploiting metadata and mastering the error detection/correction mechanisms.

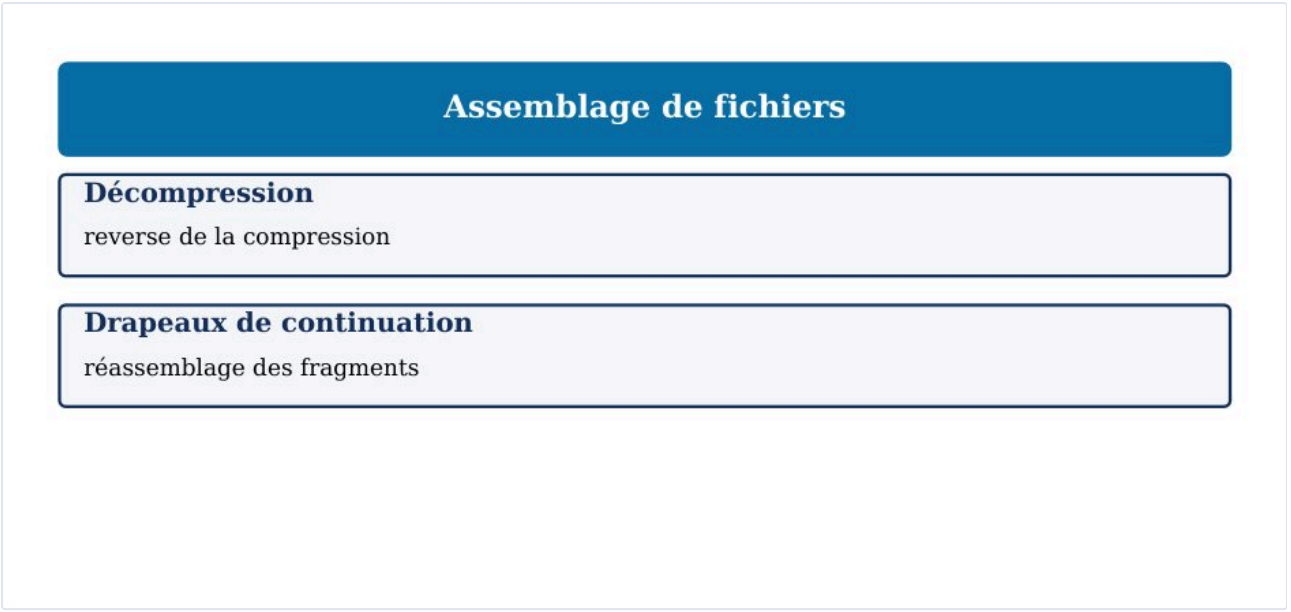


Figure 8.8 — Decompression and reassembly of packets via continuation flags.

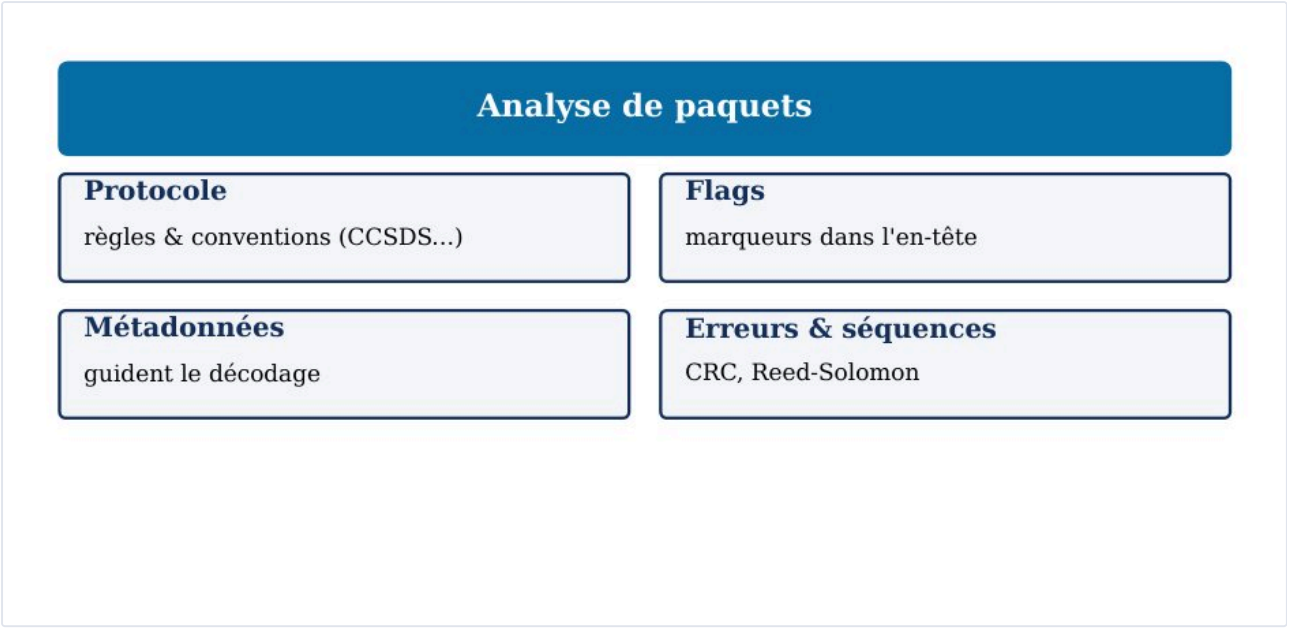


Figure 8.9 — Key elements of packet analysis (protocol, flags, metadata, sequences).

CHAPTER 10

Firmware auditing and fuzzing

10.1 Attack surface: embedded firmware

Many satellite embedded systems run firmware that can be reverse-engineered. By analysing its raw data and binaries and subjecting it to fuzzing and unit tests, you can extract information useful in an interception scenario. Satellite firmware is notably vulnerable to fuzzing.

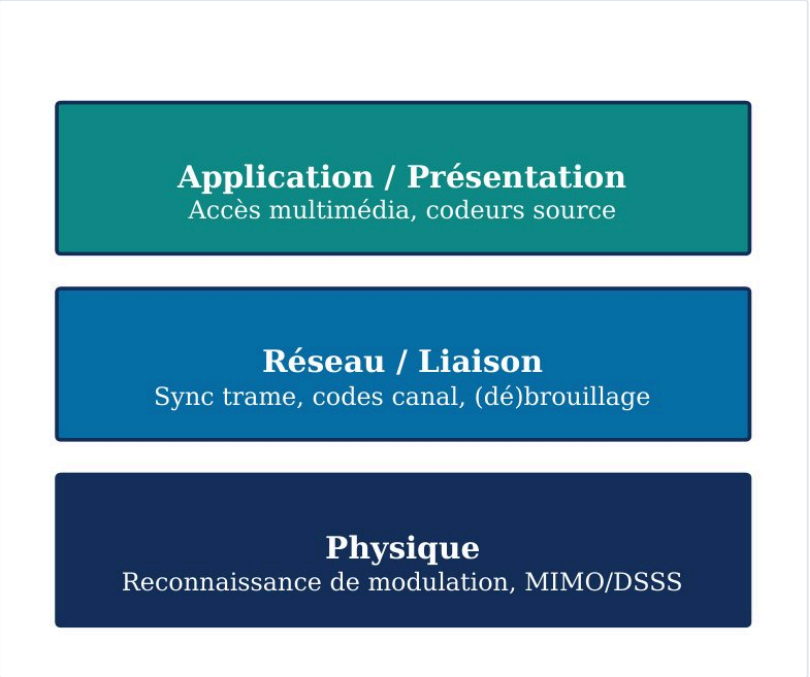


Figure 10.1 — The different levels of reverse engineering (physical, link/network, application).

10.2 Fuzzing in practice

Fuzzing means manipulating the embedded software to find vulnerabilities. It typically targets the payload data handling system, which processes the firmware and its updates. A fuzzer generates a multitude of inputs and test cases run against the extracted firmware: a full audit of the functionality, looking for errors, corruption or unexpected behaviour (crashes revealing vulnerabilities).

Methodological tips. The priority is input handling: the processing of telecommands (the satellite receives a command and runs a function). By accessing the source code or the ELF header, you identify method names, from which you infer the functionality to write targeted tests. Research work has fuzz-audited satellite firmware images and responsibly disclosed several bugs [21]. Common tools: AFL++ and boofuzz.

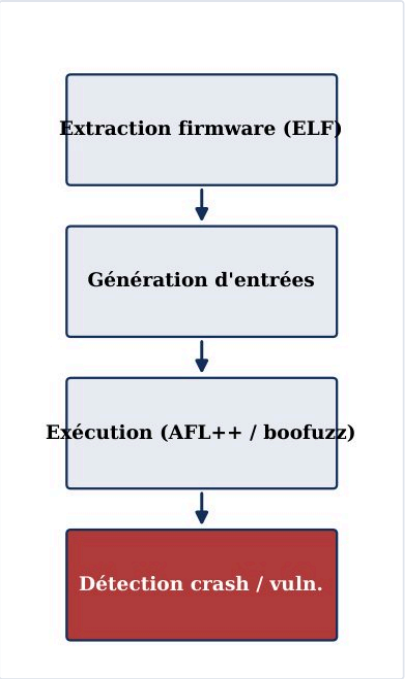


Figure 10.2 — The firmware-fuzzing principle (input handling, test cases, detection of unexpected behaviour).

CHAPTER 11

Application scenarios and summary

This chapter illustrates the full chain on three concrete scenarios, then summarises the field's tooling.

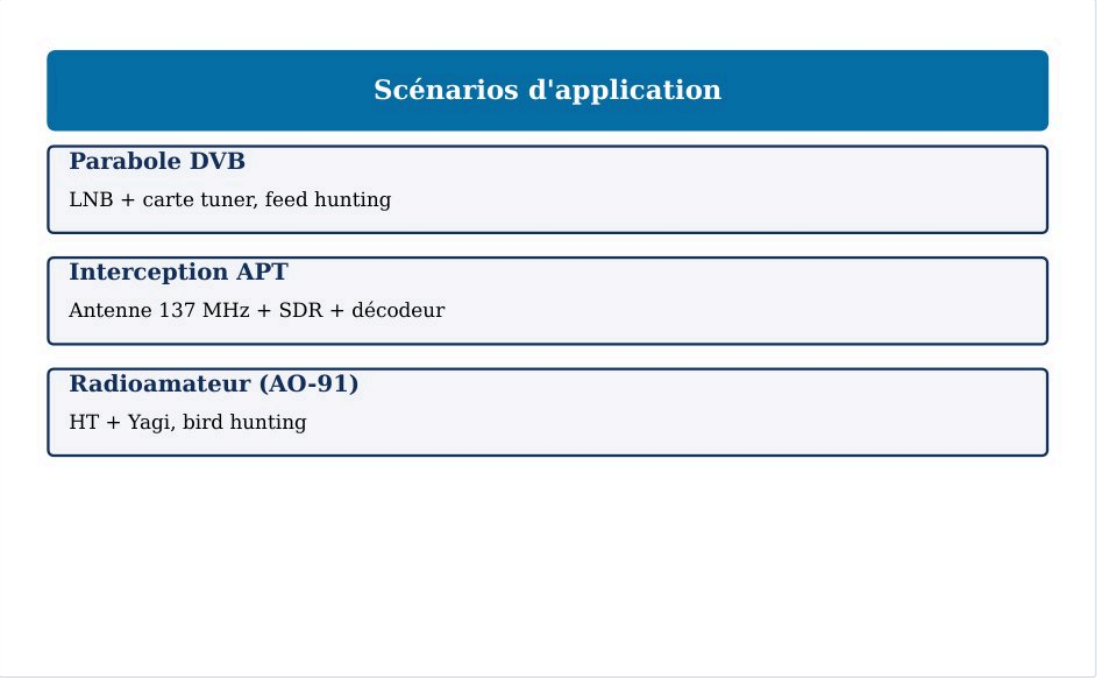


Figure 11.1 — An overview of the satellite-communication analysis scenarios.

11.1 Receiving a DVB broadcast (dish)

Receiving a DVB satellite broadcast requires a LNB (Low-Noise Block downconverter) and, as a rule, the larger the dish, the more signals it catches. A DVB tuner card (PCI) lets the computer receive the streams and record them. Feed hunting means scanning the environment to turn the streams into readable formats (binary, ASCII), using satellite-reception software that allows offline processing.

11.2 APT interception and decoding (weather)

APT interception requires four elements: an antenna tuned to 137 MHz; an SDR able to receive that band (RTL-SDR, LimeSDR, SDRplay); SDR software compensating for Doppler, in WFM mode with about 500 kHz of bandwidth; and an APT decoder (NOAA-APT, WXtoIMG or an online equivalent) to turn the captured signals into images. (See Chapter 5; reminder: this mode is now historic [8].)

11.3 Amateur-radio satellite AO-91 (Fox-1B)

Listening to amateur-radio satellites is nicknamed bird hunting (amateur satellites being called birds). AO-91 (Fox-1B) has an FM downlink at 145.960 MHz and is little affected by Doppler. You catch it with a handheld radio and a small directional antenna (Yagi or log-periodic), re-aiming it during the pass (5 to 6 passes a day). You watch for AOS and LOS, having first fetched the telemetry and pass predictions (AMSAT, N2YO).



Figure 11.2 — AO-91/Fox-1B characteristics (145.960 MHz downlink, low Doppler, Yagi listening).

11.4 Tooling summary

Function	Tool	Role
Tracking / prediction	gpredict, N2YO, Heavens-Above	Position, passes, az/el, Doppler, TLE
Dish alignment	DishPointer	Elevation, azimuth, LNB skew
Reception / SDR	RTL-SDR, HackRF, GQRX	Signal acquisition and recording
Weather decoding (APT)	SatDump, WXtoIMG	Decoding and imaging of weather satellites
RF / SDR analysis	GNU Radio, gr-dvbs2rx, URH	Signal demodulation and decoding
Packet dissection	Wireshark, tshark, dvbsnoop	Analysis and dissection (DVB-S2, GSE…)
Ground stations / telemetry	SatNOGS	Global network, dashboards, API
Binary reverse engineering	Ghidra, hexdump	Firmware and binary analysis
Fuzzing	AFL++, boofuzz	Embedded-firmware auditing

CONCLUSION

Conclusion

This survey has walked the entire satellite-communication analysis chain, from orbital reconnaissance to the reverse engineering of frames and firmware, by way of listening, decoding and packet dissection. The common thread is the field's great accessibility: an RTL-SDR, a suitable antenna and a fully open-source software stack are enough to reproduce, ethically and defensively, most of the scenarios studied.

Methodologically, three constants emerge: (1) gathering information up front (OSINT: frequencies, modulation, orbital parameters) conditions everything else; (2) knowledge of the protocols (CCSDS, DVB-S2, CSP, AX.25, APT) is indispensable for decoding; (3) reverse engineering proceeds through a reproducible pipeline (demodulation → frame decoding → demultiplexing → assembly → analysis).

Some technologies covered are heading towards obsolescence — NOAA's analogue APT has not been transmitted since 2025 [8], and WXtoIMG is no longer maintained [9] — but they remain excellent teaching material. Conversely, satellite-firmware fuzzing [21] and DVB-S2 link security remain active research areas. The chapters on attacks (firmware, DVB networks) open naturally onto the defensive side: hardening links, encryption, authenticating telecommands and segmenting subsystems.

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